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Comprehensive analysis and general economic-environmental evaluation of cooling techniques for photovoltaic panels, Part I: Passive cooling techniques

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ABSTRACT

This paper deals with the analysis of passive based cooling techniques for photovoltaic panels (PVs). A comprehensive review and evaluation of the research activities and in general studies related to the development of passive cooling techniques for PVs was obtained. A major contribution to the herein reported research study is the provision of a general economic analysis for the passive cooling options as there is a gap in present research studies related to the economic aspect of the proposed cooling techniques (the same issue was also noticed for environmental aspects). Based on the comprehensive literature review, it was found that most of the examined passive cooling options are ones with an assumed application of PCM, then air based, liquid based (water, nanofluids, etc.) and finally radiative based. A 30 kW PV plant case study was considered in order to estimate the LCOE for each considered passive cooling technique, i.e. to examine the economic aspect (where general performance data were used with respect to the obtained analysis of the passive cooling techniques). Furthermore, LCA was also carried out in order to check the environmental aspects of the considered passive cooling techniques for PVs. Finally, according to the gained results and existing technical solutions, the currently most viable passive cooling option, both from a technical and economic point of view, is the air based cooling option with Al-fins mounted on the backside surface of the PV panel. The PCM based passive cooling technique for PVs could only be an option in future terms if a significant PCM material price drop were to occur. Therefore, the future development of passive cooling techniques could be focused on the research of hybrid cooling options. The hybrid passive cooling option assumes a mix of passive cooling techniques. Finally, the advantage of each cooling technique could be efficiently utilized in that manner.

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1. Introduction

Nowadays, photovoltaic technology is widely used in different applications, [1] for renewable electricity production. Without the more intense application of photovoltaic technologies and other renewables as well as the development of novel and alternative renewable based energy concepts, [2,3] we would not be able to reach the general targeted goals related to the restriction of energy consumption and reduction of harmful impacts to the environment. However, and despite a significant reduction in capital investment required, the overall initial investment cost for PV systems is an issue, especially for smaller systems when considering their rather modest energy conversion efficiency. The average initial investment of a PV system for residential applications usually ranges from about 2.6 USD/W to 3.4 USD/W, while for large systems the overall investment can range from 2.0 USD/W to 2.3 USD/W on average, [4]. Furthermore, if we analyze the cost structure of the overall investment related to the PV system, the highest share falls on the PV panel itself (about 0.64 USD/W, [4]). In this line of approach, improvements in PV technology is a crucial factor in order to boost their market propagation, in accordance with the main targets of the Paris climate agreement and the EU goals for 2030, [5,6]. Currently, the older Silicon (Si) based photovoltaic technology (usually in Si-poly or Si-mono variant) accounts for the largest market share, with an average efficiency usually ranging from 10% to 15%. Although, the Si-mono variant has higher efficiency than the Si-poly one, the Si-poly PV technology has become more popular in recent years due to a lower overall investment cost, by about 20–30% on average, as well as due to reasonable

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Nomenclature

AC	Total life cycle cost, €	*n*	Amortization period, years
CRF	Capital recovery factor	OM	Operation and maintenance cost, €/year
EO	Average annual overall energy output from the hybrid energy system, kW h/year	*p*	Interest rate, % P.a.
IC	Installation cost (overall investment), €		

efficiency when compared to other PV technologies. It is also important to stress that the Si-poly technology is more sensitive to the PV panel's operating temperature, when compared to Si-mono, and which affects the overall PV panel efficiency. Other market available PV technologies, like ones from amorphous silicon, thin film (CdTe) or CIS/CIGS are less represented on the market due to a significantly lower energy conversion efficiency (about 6–12%, depending from the technology) and due to other general issues. Significant research efforts are being implemented in order to find novel PV technologies with a higher efficiency and to achieve economically more viable PV technologies. However, we are dealing with technologies that are not ready for a wide market implementation and that are currently under intense research activity, so it is more likely that Si-poly or Si-mono PV technologies will further have the highest market share in the near future. Focusing on research efforts that try to improve existing market available Si-based technologies in order for them to become more attractive and more economically reasonable are therefore more worthwhile.

It is well known that the operating temperature of PV panels strongly affects the already modest PV technology efficiency, where about 0.25%/°C up to 0.5%/°C is the rate of expected PV panel efficiency degradation caused by the rise of PV panel operating temperatures, [7]. Thus, proper cooling techniques for PV panels could ensure additional and desirable increases in PV panel energy conversion efficiency. Another favorable aspect associated with the implementation of cooling techniques for PVs is the prolonged premature degradation of PV panel efficiency (increased lifetime). According to some authors, the lifetime of PVs can be prolonged from about usually 25–30 years up to 48 years [8], by applying the specific cooling techniques for PVs. Furthermore, PV cleaning issues can also be resolved by applying certain cooling techniques and hence achieving an additional increase in annual average delivered electricity yield.

For these aforementioned reasons, significant research efforts were focused on during the last two decades by investigating different cooling techniques which can be in general divided into passive and active. The major concern related to the existing research findings is related to the economic and environmental aspects of the proposed cooling techniques for PVs. Several review research studies were also reported, however, they focused on specific PV applications (solar concentrator photovoltaics – CPV, solar thermal–PV/T, photovoltaics for building applications, i.e. building integrated photovoltaics–BIPV, etc.). For example in paper [9], a critical review of cooling techniques for photovoltaics was obtained for solar concentrator systems (CPV). Namely, different CPV systems have been analyzed from a base application aspect and finally grouped according to their geometry. Furthermore, both cooling and passive cooling techniques are summarized and each cooling technique was evaluated on a few criteria such as; heated area, pump power, pressure drop, mass flow rate and thermal resistance. It is important to emphasize that comparisons were reported both for theoretical and experimental studies. The study results [9] imply that micro channels or impinging jets are the best solution for the cooling of solar CPV systems with concentration levels higher than 150 suns. Paper [10] provided a review of thermal management techniques for PV systems. Namely, different cooling techniques for photovoltaics (natural, forced, hydraulic cooling of PV/T systems, water impingement cooling, Heat pipe cooling and PCM) have been addressed. The main outcome of the study [10] provided a summary of general data related to PV system type, and finally data related to PV panel operating temperature. Authors found that naturally ventilated PV systems will work in a range of operating temperatures between 50 °C and 70 °C and a forced one between 20 °C and 30 °C. It also emphasized that the De-ionized liquid immersion method can reduce operating temperatures by 30 °C to even 45 °C in the case of CPV systems. Finally, PCM systems turned out to be viable cooling options according to the authors. However, authors did not analyze the economic aspects of the proposed cooling techniques.

Different cooling techniques for PVs were addressed in [11], i.e. for hybrid photovoltaic systems, water and liquid based PVT systems, refrigerant based PVT systems, heat pipe PVT systems, PCM based PVT collectors and finally thermoelectric cooling (almost all of the cooling techniques for PVs were considered). The review paper ended by summarizing the advantages and disadvantages of each cooling technique mentioned above. Important parameters for each analyzed cooling technique, like for example the PV technology type, improvement of electrical efficiency, thermal efficiency, overall exergy efficiency and reported PV operating temperature were specified. Finally, authors addressed that water based cooling systems within PVT configuration are the most promising ones and have the most specific advantages. A numerical and experimental study related to the novel PV/T system was elaborated in [12]. The authors reported an average increase in thermal efficiency by about 41.9% and about 9.4% in electrical. The system was investigated for typical Chinese climate conditions and it is suitable for operation in cold regions without freezing issues. The optimization and design for the PV/T heat-pipe system with a PCM was addressed in [13] through a developed numerical model and for Beirut climate conditions. An optimal system was found with improved general efficiency. The theoretical aspect of the considered PV/T system with an integrated compound parabolic concentrator was elaborated in [14] (where the developed model allowed a detailed performance analysis). Different aspects were addressed in [15] and related to the PV/T configuration (a specific simulation model was developed to obtain analysis).

Various cooling techniques were addressed in [16] as for example the use of PCM, water passive and active, evaporation, heat-pipe and finally air forced techniques. The analysis was finalized with the comparison of different cooling techniques based on the achieved effective increase in PV panel peak output. The comparison was therefore only obtained for those studies which reported a net increase in PV panel output, after which power losses are taken into account and related to the considered cooling technique. According to the authors, the best results were achieved for water forced cooling techniques for photovoltaics and the most promising systems are the PV/T systems where an efficient usage of rejected heat can significantly contribute to the economic viability of the mentioned systems.

The main objective of the herein elaborated research study was to obtain a comprehensive analysis and evaluation of passive cooling techniques for photovoltaics, i.e. to compare general technical and performance features. Based on the herein provided comprehensive review of passive cooling techniques, a 30 kW PV system case study was considered in order to calculate the general LCOE for each examined cooling option and hence determine their economic aspect as an important feature that is missing in most research findings published thus far. Finally, environmental aspect analysis was also conducted for the considered passive cooling technique through obtained LCA analysis.

2. Comprehensive analysis of passive cooling techniques for photovoltaics

This section addresses relevant references that examined different passive cooling techniques for PVs in the last two decades. The main point of this section was to provide a focused review and to set a reasonable base for obtaining crucial economic analysis later.

In paper [17], a specific profile surface was developed to enable illumination for different concentration rates (photovoltaic system with solar concentrator). However, the same work proposed a passive cooling method based on thermosyphons, i.e. passive heat pipe cooling system. Namely, two heat exchangers were assumed, Fig. 1, where the lower heat exchanger was an evaporator (filled with refrigerant R-11 or R-22) and the upper one was a condenser. The circulation of refrigerant in the system was due to gravity, i.e. non forced circulation. The proposed cooling technique was able to reduce the operating temperature from 84 °C to 46 °C, with a reported increase in power output from 10.6 W up to 20.6 W. The reported efficiency of the photovoltaic panel was around 10% at 46 °C.

Paper [18] dealt with the application of phase change materials (PCMs) for building integrated photovoltaic systems. Thermal performance was analyzed for different fin arrangements, Fig. 2. Different fin spacing was used, i.e. from 4 mm to 20 mm, and fin effect on temperature control was examined, with respect to different widths, fin types and spaces between the fins. The authors reported that the proposed PV/PCM configurations can reduce PV panel operating temperature by more than 30 °C (for solar irradiation levels of 750 W/m^2 and with an ambient temperature of around 23 °C). Solid-liquid (RT25) and granular PCM materials (GR40) were tested and it was found that the granulated material is not effective as a solid-liquid PCM material with respect to thermal control (the solid granules in the GR40 material reduce bulk thermal conductivity). The main outcome of the considered paper was a comparison of electrical conversion efficiency for the proposed PV/PCM concept with a base case flat aluminum plate. For the proposed configuration, fin height and space between fins have a crucial role regarding thermal response (in their study, the optimal fin height was found to be around 40 mm for UK climate conditions with a fin spacing of 4 mm and for a PV building integrated system in a horizontal position).

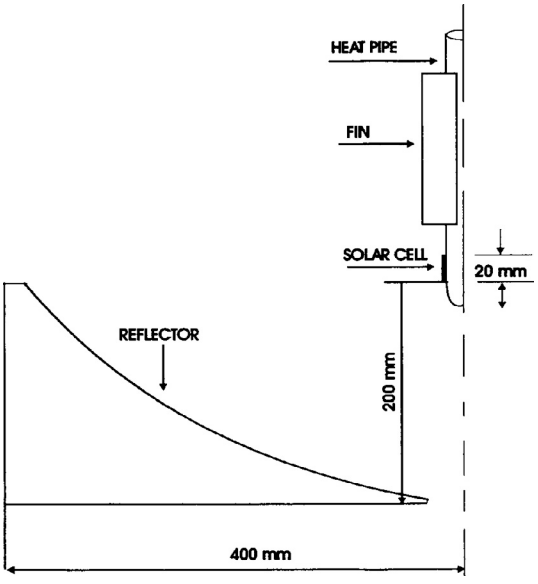


Fig. 1. Simplified sketch of concept, [17].

A numerical study related to rooftop integrated photovoltaics was obtained in [19] and a developed model was tested for different channel geometries (channel spacing and channel length, Fig. 3). The channel length was variated from 1 to 10 m, channel spacing from 0.02 m to 0.2 m, channel orientation from 90° to 30° and channel cooling rates from 0 to 500 W/m^2 (with assumed ambient temperature of about 27 °C). The effect of the main geometric parameters on PV panel electrical efficiency was presented in Fig. 4 (for solar irradiation levels of 500 W/^{2} and 1000 W/m^2). It was found that an increase in channel spacing leads to an increase in efficiency, with a higher increase of efficiency for higher solar irradiation levels (for 0.2 m of channel spacing the increase in electrical efficiency is 0.3% for 500 W/m^2 and 0.5% for 1000 W/m^2). The estimated reduction of PV panel operating temperature was expected to be between 10 °C and 20 °C with an absolute efficiency increase from 1% to 2%.

In paper [20], authors proposed a cooling method for PVs by water submersion. They obtained four month experiments with silicon PV panels that were submerged in still water. A referent PV panel exposed to air was also used to compare results. The water depth was ranged from 4 cm to 40 cm and specific results in relation to relative PV panel efficiency are presented in Fig. 5. The results were obtained for different silicon based PV market available technologies. According to the reported results, the referent PV panels had average operating temperatures between 70 °C and 80 °C, while the submerged PV panels had average temperatures at around 30 °C. The previous implied an efficiency increase from 13% to 15.5%, therefore the maximal increase in efficiency being around 20% (the average increase in efficiency was around 11% for the case of 4 cm water layer depth). Peak power output was increased from 19 W to 29 W for a solar irradiation level of around 800 W/m^2. However, it was also found that by increasing water layer depth, efficiency is degraded and not increased (at 40 cm of water layer, the reduction in efficiency was found to be −23%). The reason for the previous decrease in PV panel efficiency is due to significant sunlight beam reflection and refraction in the water. It is obvious from their reported research that optimal water depth would be somewhere between 2 cm and 4 cm. The authors have continued their research and are currently dealing with floating photovoltaic plants.

An evaluation of PCM materials for applications in building integrated photovoltaic systems was obtained in [21] (RT 20, Eutectic mixture of capric–lauric acid (C–L), Eutectic mixture of capric–palmitic acid (C–P), Pure salt hydrate (CaCl2_6H2O) and SP22). Different experimental configurations were considered (Fig. 6) and experiments were obtained in laboratory conditions, using a solar simulator (three levels of solar irradiation were used and ranged from 500 W/m^2 to 1000 W/m^2). According to their findings, it was concluded that thermal PCM regulation performance depends from the PCM thermal mass as well as thermal conductivity of the whole system (where thermal conductivity has more significant influence for lower solar irradiation levels). The achieved temperature reduction ranged from 10 °C up to 18 °C. Finally, it was concluded that it is necessary to ensure a

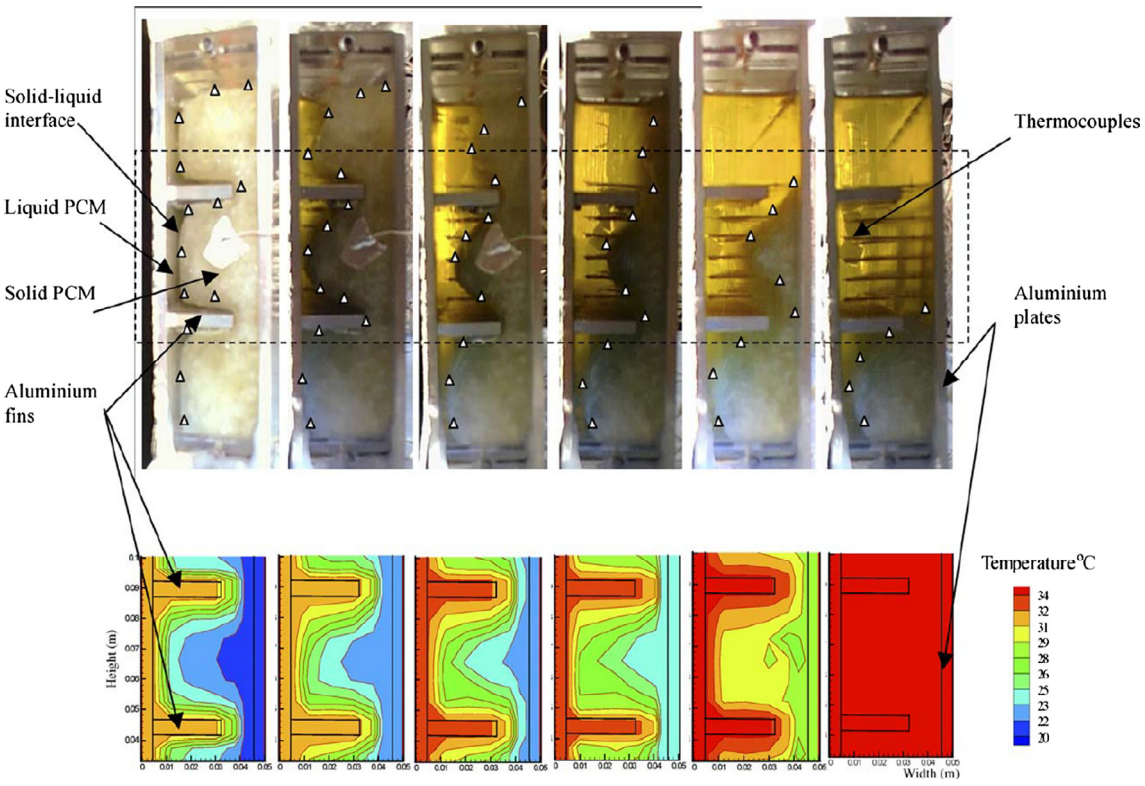


Fig. 2. Different fin arrangements and thermal response, [18].

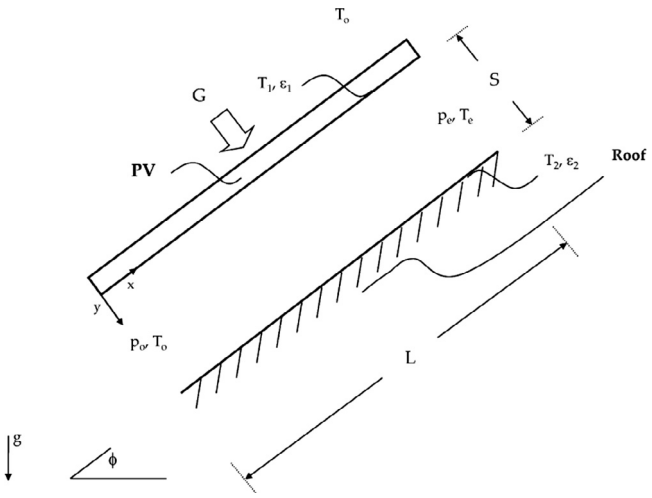


Fig. 3. Base geometry analyzed in numerical experiments, [19].

further reduction in PV panel operating temperature in order to ensure PCM material economic viability for the assumed building application (the possibility of using thermal conductive metal fins had been addressed). The main advantages and disadvantages of different used thermal management techniques had also been addressed.

In paper [22], authors obtained an experiment related to the implementation of a passive cooling technique. Namely, an aluminum heat sink was attached on the backside surface of the PV panel in order to increase heat rejection, Fig. 7. Experiments were conducted in laboratory conditions using a solar simulator and the achieved increase in peak PV panel output was about 24% (20% increase on average) for a solar irradiation level of 800 W/m^2, with an increase in electrical efficiency from 3.96% to 4.76% (solar irradiation levels ranged from 200 to 800 W/m^2 in general, and surrounding air temperatures also ranged from 15 °C up to 35 °C). An average increase in electrical efficiency ranged from 0.64% up to 0.8% depending from the applied solar irradiation level. Authors also obtained a theoretical analysis and compared it with the

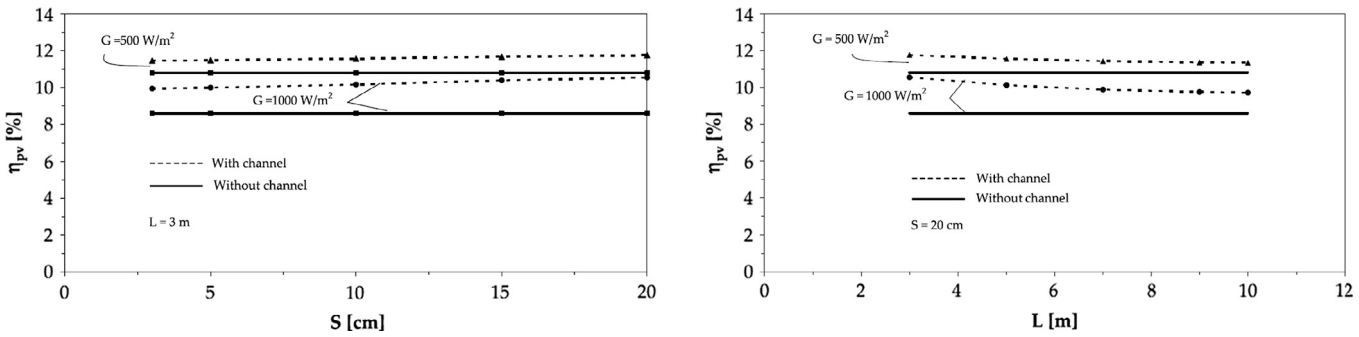


Fig. 4. PV conversion efficiency as function of channel spacing (S) and length (L), [19].

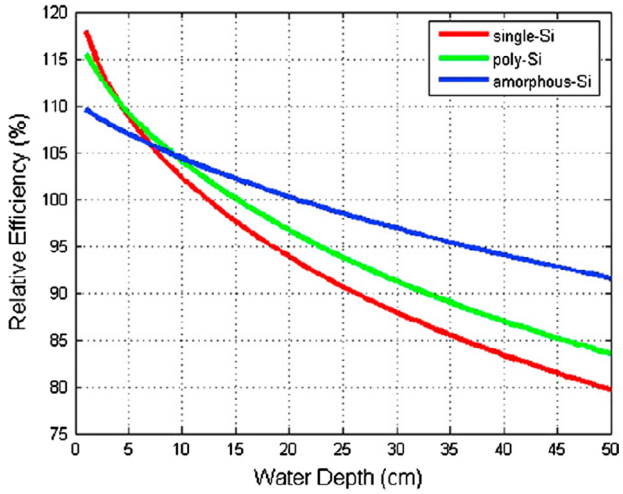


Fig. 5. Relative PV efficiency as variation of water depth, [20].

experimental results; however, a small size PV cell had been (authors did not specify the nominal power output of the examined PV cell). For the examined configuration, it had been found that maximal cooling rate was achieved for an applied illumination of 600 W/m^2. It had also been found that for low solar irradiation levels, passive cooling becomes inefficient due to the small temperature difference between the air and fins (around 1 °C).

Experimental as well as theoretical PCM material evaluation, i.e. effects related to heat convection and segregation issues were obtained in [23] for PCM material application in building integrated photovoltaics (three types of paraffin based PCM materials were used). An experimental configuration is presented in Fig. 8, and tests were done under laboratory conditions where the simulated solar irradiation level was 750 W/m^2 and the surrounding air temperature was 19 °C. The maximal reached plate temperature was 63 °C and by applying the PCM material, the temperature of the PV panels was reduced to 42 °C. The fin spacing was also variated, and results were compared with the module without a fin, with respect to the thermal management of the PV/PCM system. It was found that fins in general reduce temperature rise and thermal stratification. However, fins reduce the period over which temperature is controlled and by that they in general increase the weight of the overall system.

Paper [24] also deals with the application of PCM materials to regulate the BIPV system thermal load. A numerical simulation model was developed for single PCM applications to predict thermal performance in PV/PCM systems with triangular cells, Fig. 9. For simulation purposes, UK conditions (June SE UK) were considered, however, for circumstances when the solar irradiation level was above 120 W/m^2. Two types of PCM materials had been considered and coupled with two different shapes of fins (triangular and half circular). By using PCM materials, PV panel electricity production was increased from 1.4 kW h/day m^2 to around 1.8 kW h/day m^2 and the PV/PCM system temperature remained under 30 °C, while the reference plate was over 50 °C. The combination with triangle cells achieved the highest decrease in PV/PCM system temperature during daily operations.

PCM material, i.e. metal wax-composite PCM matrix was used for the passive cooling of the photovoltaic module with V-trough, Fig. 10, [25]. The photovoltaic tested module was 10 W in nominal power output and the system was filled with 5.5 kg of paraffin wax that was exposed to a tungsten halogen lamp of 1000 W in rated power with an inclination of 30° (as presented in Fig. 10 and where a referent PV panel was used to compare effects related to the application of a PCM material). Besides laboratory experiments, outdoor experiments were also performed as well as numerical modelling (with reasonable experimental data matching). In laboratory conditions, the PV panels were exposed to a solar irradiation level of 2300 W/m^2, where the referent PV panel temperature was around 90 °C. However, the PV panel that used a PCM material had 15–20 °C of lower operating temperatures than the referent PV panel, i.e. between 65 °C and 68 °C. Outdoor experiments were obtained for clear sunny days with a resulting incident solar irradiation of 1982 W/m^2. The referent PV panel had operating temperatures between 78 °C to 80 °C while the PCM-controlled PV panel recorded 64–65 °C (therefore, the achieved temperature reduction was around 15 °C). The PV panel power output increase was around 55% and the increase in delivered electricity was from 0.0486 kW h to 0.0754 kW h (electrical efficiency was increased from 5.40% to 5.91%). The authors stressed the issue of possible chemical degradation, i.e. the importance of applied metal as well as sealing issues.

Peltier elements for photovoltaic cell cooling were proposed in [26]. A simulation model was developed using MATLAB as well as genetic algorithms in order to optimize the system (i.e. to find the optimal value for the input current that will ensure maximal PV panel power output). The base concept is presented in Fig. 11 and was based on a simulation model that was developed. Different solar irradiation levels were applied and the optimal level of solar irradiation was of 1000 W where the maximal reduction of operating PV panel temperature was around 16 °C (but it was found that in the considered case, the cooling system requires more power than the PV system is capable to produce). Therefore, the optimal reduction in the PV panel operating temperature was found to be below 8 °C, which leads to a modest expected increase in PV panel efficiency.

Authors in [27] proposed a passive cooling technique for PV panels using cotton wick structures that were mounted on the backside surface of PV panels, Fig. 12. Different fluids had been tested in combination with the cotton (water and nanofluid water, i.e. CuO/water and Al_2O_3/water with 0.1% volume concentration).

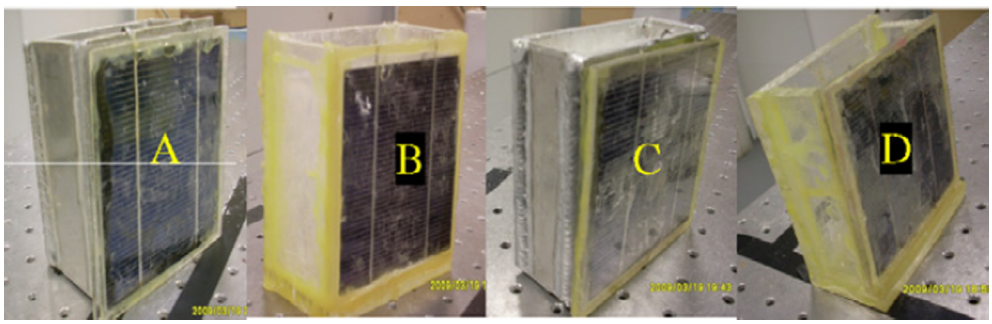


Fig. 6. Experimentally tested configurations, [21].

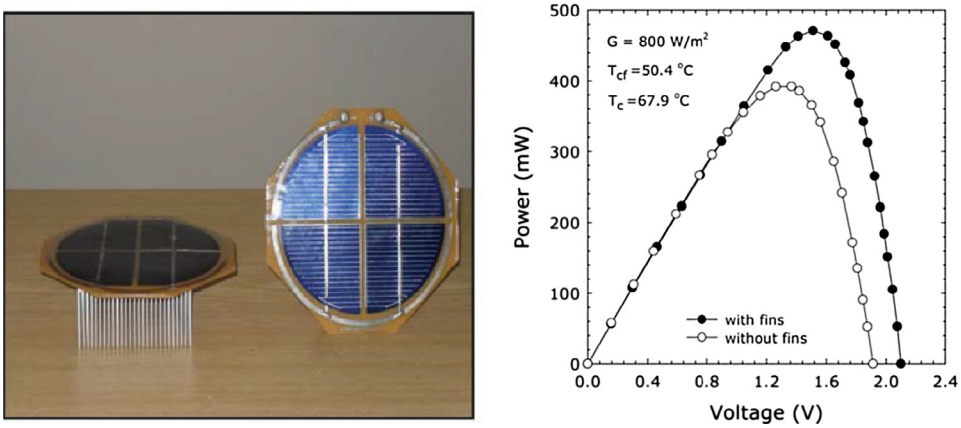


Fig. 7. Characteristic aluminum fins attached to photovoltaic cell and typical increase in PV panel power output, [22].

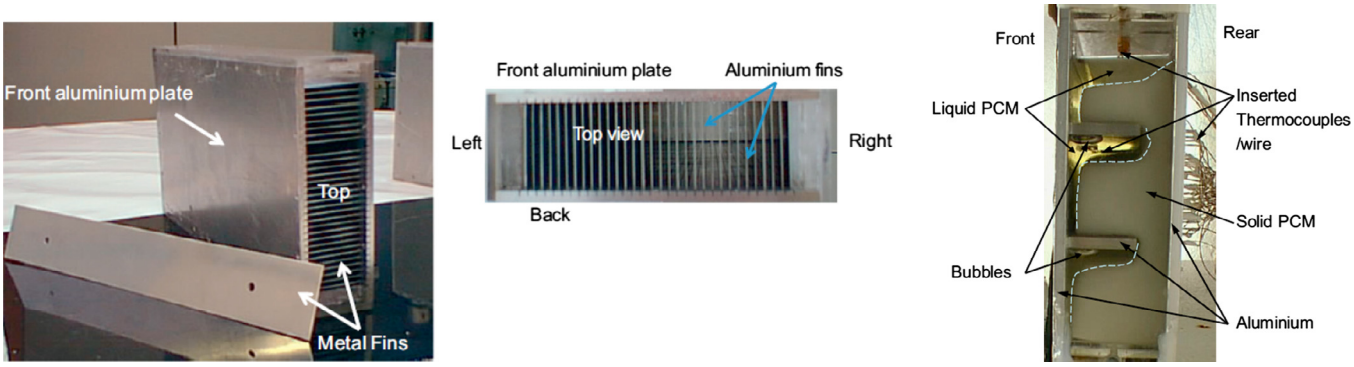


Fig. 8. PV/PCM system, i.e. experimental configuration, [23].

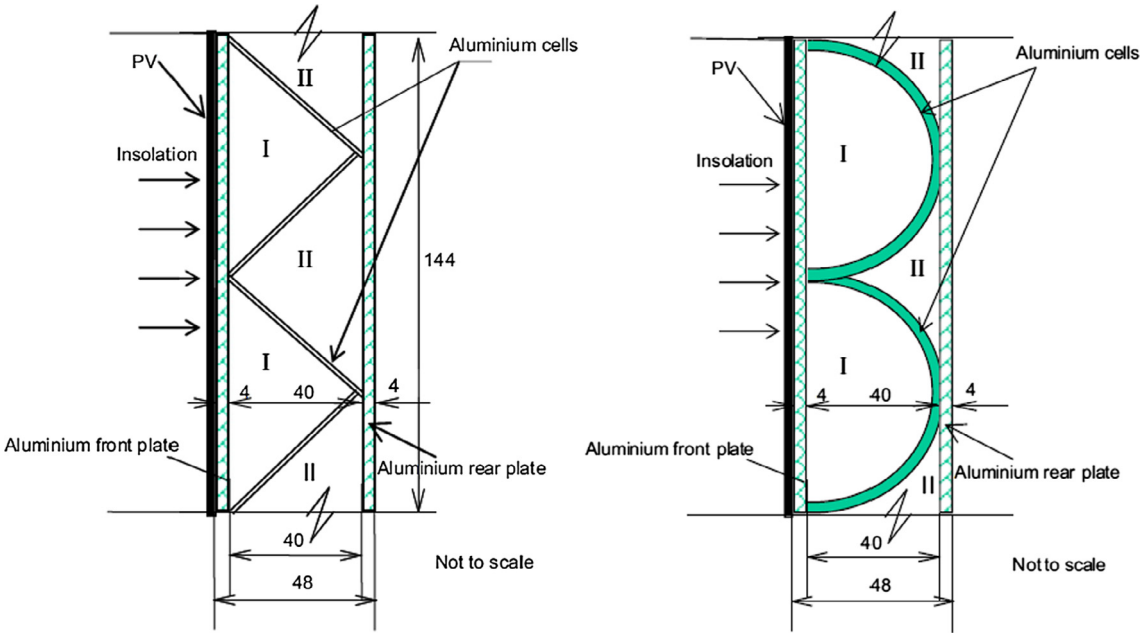


Fig. 9. Basic sketch of examined PV/PCM system, [24].

The main idea was to utilize the capillary action of the wick structures in order to enhance the heat rejection from the PV panels to the surroundings. Solar irradiation ranged during the experiments from 0.1 MJ/m^2 to around 0.5 MJ/m^{2,} with average surrounding air temperatures of 37 °C. The combination of cotton wick with water reduced the PV panel operating temperature from 65 °C (maximal value) to about 45 °C. It was also found that the previous combination leads to a more uniform distribution of PV panel operating

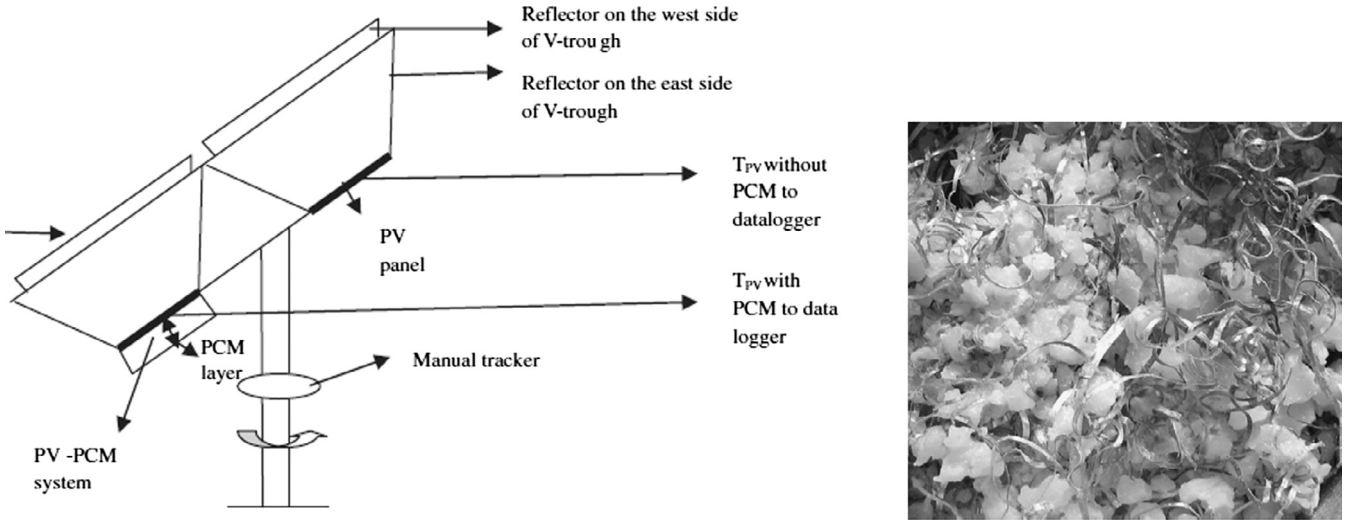


Fig. 10. Scheme of experimental setup and passive coolant, i.e. paraffin wax in metal-embedded PCM matrix, [25].

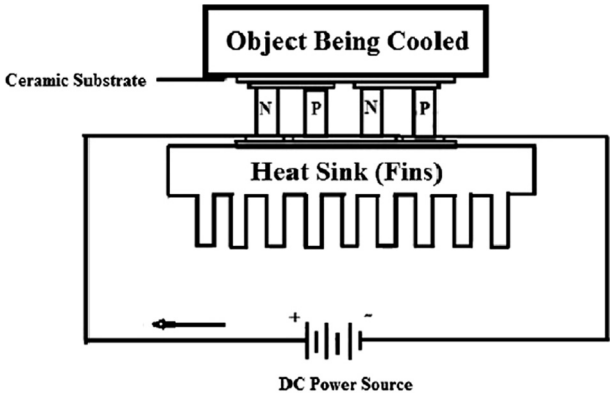


Fig. 11. Schematic overview of proposed systems for thermoelectric cooling, [26].

temperature during day time. With the application of nanofluids, the maximal operating temperature of the PV panel was around 59 °C for CuO/water and 54 °C for Al_2O_3/water nanofluid. The highest increase in PV panel power output was achieved for the case of wick structures in combination with water in the amount of 47.5 W, while the non-cooled PV panel delivered around 41 W, therefore, the obtained increase was 15.8% (while for nanofluids the increase was 8.8%). The average electrical efficiency without cooling arrangements was around 9% and the highest one was again achieved for the combination of wick structures with water, in the amount of 10.4% (maximal value). In the case where nanofluids were applied, the PV panel electrical efficiency ranged from 9.4% to 9.8%.

The liquid-immersion method was applied directly on the PV cells to enable efficient thermal management in [28]. Different types of fluids were examined with variable liquid layers (from 1.5 mm to 9.0 mm), i.e. De-ionized water (DI), isopropyl alcohol (IPA), dimethyl silicon oil and ethyl acetate were applied. The experimental setup is presented in Fig. 13 and the PV cells (mini cells) were exposed under 30 suns in laboratory conditions where the temperature was controlled to about 25 °C. The highest increase in panel efficiency was obtained for isopropyl alcohol, 15.2% and the lowest one for De-ionized water, 8.5% (for the lowest liquid layer of 1.5 mm in both cases). Long-term cell stability was also examined for 170 days in duration and for different sun levels from 10 to 30 suns, and for 9 mm thick liquid layers. However, for

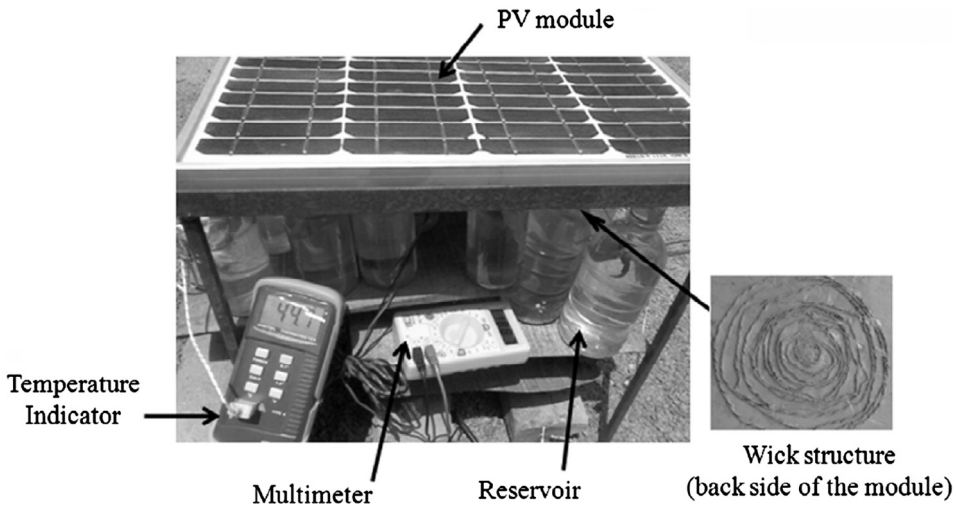


Fig. 12. Photo of the characteristic experimental setup, [27].

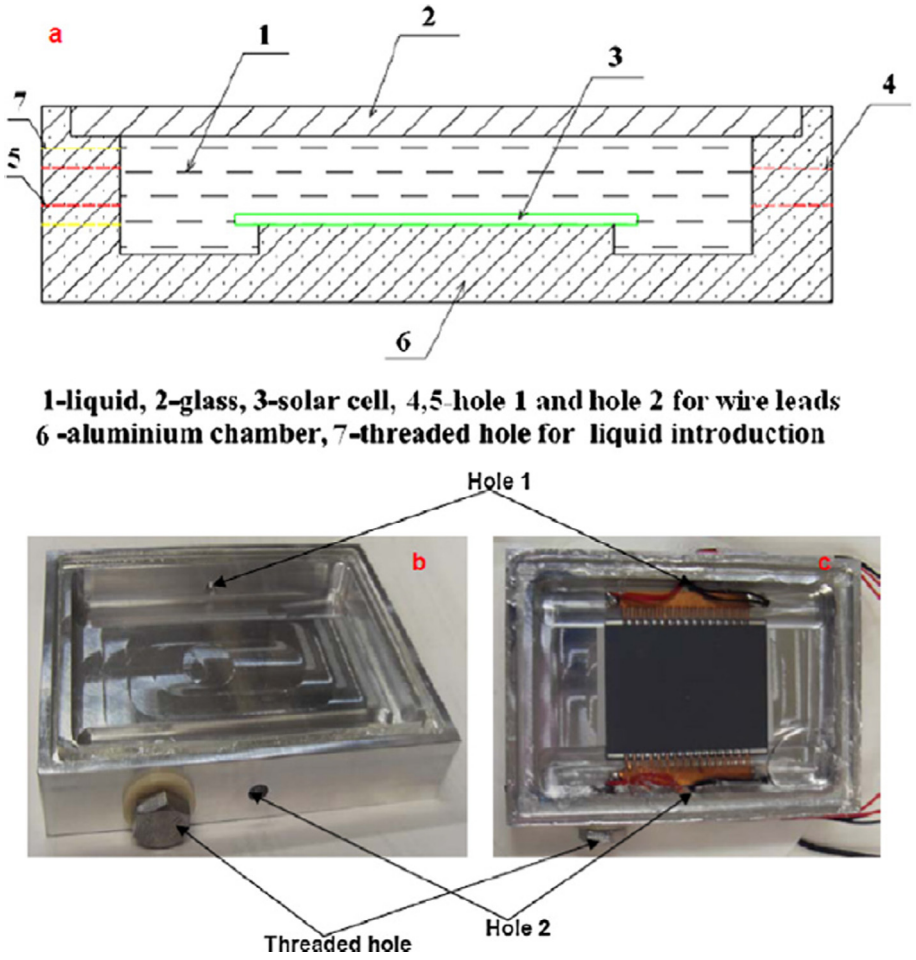


Fig. 13. Chamber and mini PV module, [28].

larger fluid thickness levels, efficiency was additionally degraded and the highest decrease was for a 9 mm thick liquid layer. In the case of long-term stability, all the applied fluids showed stability, with an insignificant impact on electrical efficiency; however, DI water caused long term efficiency degradation.

Paper [29] dealt with CFD simulations related to the application of PCM materials as a passive cooling technique for PV panels. The focus of the study was the development of a CFD model and the modelling of complex heat and mass processes. Typical PCM configurations (application of the PCM material on the backside surface of the PV panel coupled with fins) were numerically examined and experimentally validated, Fig. 14. The main idea was to determine conditions to maintain PV panel temperature with the surrounding temperature. For the purpose of numerical simulations, the applied solar irradiation level was 1000 W/m^2. Different configurations had been simulated, i.e. different dimensions of the PCM layer, as well as configurations with and without fins. By applying the PCM material, it was found that major decreases in PV panel operating temperature occur in the first hour and range from 55% to 57% (depending from the tested configuration). After two hours, the temperature is reduced from 21% up to 36.6%. The average operating temperature of the PV panel that was maintained during 80 min was 40 °C, while for the non-cooled PV panel the same temperature was reached in just 5 min. In general, the maximal temperature reduction was from around 85 °C to 38 °C (the PCM with the highest thickness and fins included) where it was found that the temperature is proportional to PCM width.

The application of a PCM material was considered in the reported research [30] and a specific experimental setup was presented in Fig. 15. According to Fig. 15, the system consisted of two PV panels of 10 W in nominal PV power output, i.e. one referent and the second one equipped with aluminum rectangular tubes that had been filled with the PCM material (around 1.0 kg of yellow petroleum jelly with melting temperatures from 42 °C to 55 °C). The experimental configuration was tested horizontally and with an inclination for Indonesian climate conditions (Bandung). The maximal recorded solar irradiance was 1120 W/m^2 with a surrounding temperature of around 25 °C. For the maximal solar irradiation level, the referent PV panel had averaged an operating temperature of about 59.4 °C, while the one with the PCM had a temperature of around 55 °C, and the maximal recorded decrease in the PV panel operating temperature was 4.4 °C. For the on-stand PV experiment, the maximal recorded decrease in PV operating temperature was 2.4 °C, while the average was 0.4 °C (solar irradiation 1049 W/m^2). The average increase in PV panel efficiency was 21.2% for the on-roof PV configuration and 6% for the on-stand PV configuration.

A passive evaporative cooling technique for PV panels was proposed in [31] and tested under climate conditions for the United Arab Emirates (Sharjah). The experimental set-up as well as the specific construction of the cooling elements was presented in Fig. 16, where 10 W of monocrystalline PV panels were used in the experiment. Besides the experimental approach, an evaporative cooling technique was also elaborated in detail from its theoretical point of view. The cooling element was obtained from

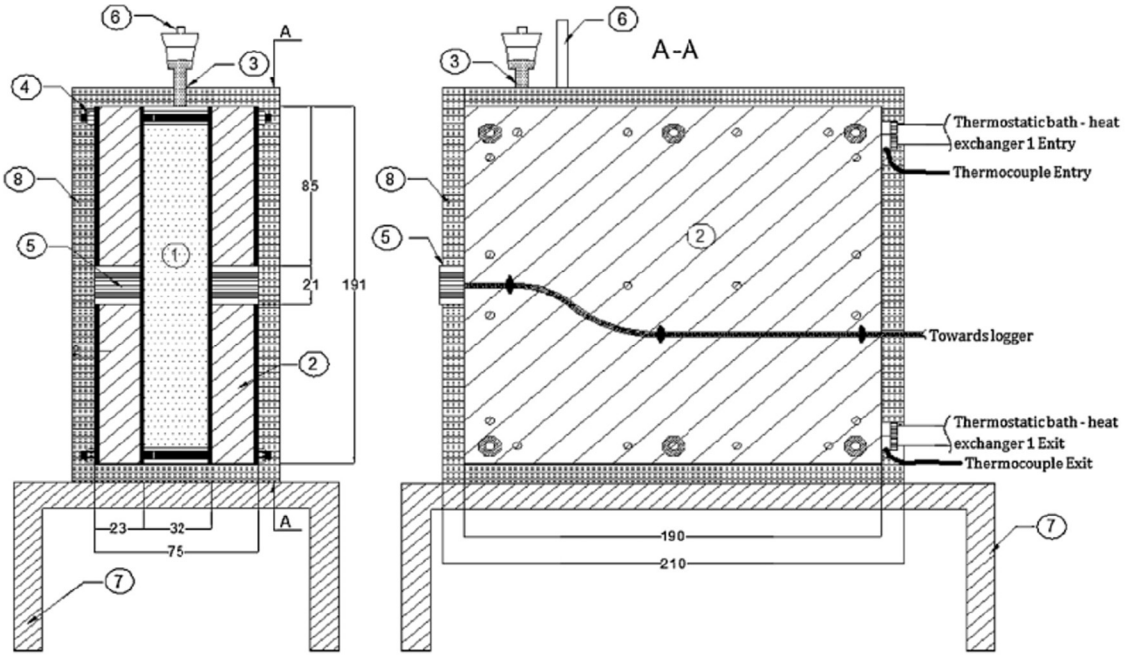


Fig. 14. Schematic of experimental setup with PCM material, [29].

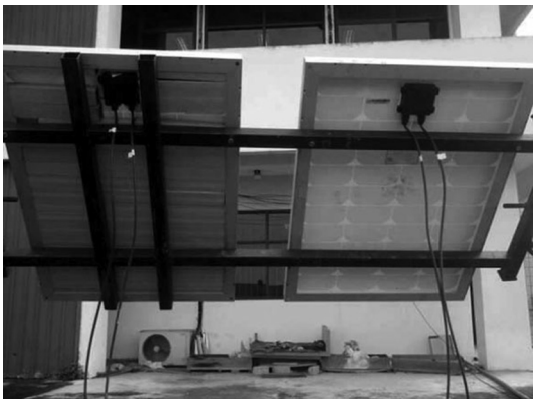


Fig. 15. Specific experimental setup for passive PCM based cooling technique, [30].

copper substrate (perforated with 6 mm holes in arrays), i.e. 1 mm thick sandwich layer between two layers of Kera clay-water based synthetic clay (right photo, Fig. 16). For a solar irradiation level of about 900 W/m^2 it was found that the operating temperature for the PV module with no cooling was 85 °C and with the proposed cooling system the operating temperature was 45 °C. The average increase in PV power output ranged from 14.2% to 19.1% and an increase in electrical efficiency was not reported.

An investigation of various PCM materials was obtained in [32] by using differential scanning calorimetry and a temperature history method (THM). The main goal of the research was to examine the thermo-physical properties of the considered PCM materials and their response to PV panel temperature regulation for specific operating conditions. Three general types (classes) of PCM materials were analyzed; paraffin waxes, salt hydrates and mixtures of fatty acids (RT20, SP224A, CaCl2-6H_2O, CL and CP). Each considered PCM material was heated from 10 °C to 60 °C at an identical heat rate in the duration of 5 min. The THM method was applied in the way that a melted sample of 25 g for each PCM material was placed in the test tubes that were heated at 40 °C and exposed to the cooled environment (5 °C) for a few minutes. The main outcome of the research study was the relationship between the



Fig. 16. Experimental setup and cooling elements for evaporative cooling, [31].

examined PCM materials and their effect on temperature regulation. It was also additionally found that $CaCl_2$-$6H_2O$ and SP22 PCM materials have higher under-cooling during summer nights when compared to other examined materials. Further, it was also found that the SP22 material solidifies over a high temperature range and that the CP PCM material is more suitable for PV temperature control than the CL.

A water immersion method (distilled water) was considered as a cooling option for the PV in the study [33]. A polycrystalline PV panel (0.119 W) was submerged in water and tested under Iraqi climate conditions (Dohuk), with an average solar irradiation level of 700 W/m^2 (with a surrounding air temperature of around 36 °C and with a wind speed of around 1.0 m/s). Different depths of water layers had been variated (from 1 cm to 8 cm), and the PV response was presented in Fig. 17 as the function of different examined water depths. The maximal obtained PV panel efficiency was found to be for a 6 cm thick water level in the amount of 22%. The average increase in PV panel efficiency was about 11% for a 6 cm thick water layer (water temperatures ranged between 28 °C and 30 °C). A referent PV panel for comparison of results was not provided.

Paper [34] analyzed different title angles for BIPVs with respect to the PV panel operating temperature. A title angle optimization procedure was obtained (title angles ranged from 0° to 60°) in order to maintain, as much as possible, a lower PV panel temperature (different cost effective cooling arrangements were considered). The analysis was conducted by using CFD modelling and the gained results were compared with available experimental data, where the solar irradiation levels ranged from 200 W/m^2 to 1000 W/m^2. It was found that minimal operating temperatures were obtained for a title angle of 15° and 60° for maximal operating temperatures. Different PV technologies were considered and the best obtained result was for the TF-Si based PVs, Fig. 18 (case of optimal tilt angle that was compared to the case of 0°).

A global analysis related to the application of PCMs for the cooling of PVs was obtained in [35] for different regions of the World (Mexico, Eastern Africa, Central and South America, Europe, etc.). Simulations were obtained for the model presented in Fig. 19. A variation of PCM melting temperatures ranging from 0 °C to 50 °C was obtained in order to detect the optimal melting temperature for each considered geographical location. It was generally found that higher average surrounding temperatures will cause higher melting temperatures for PCMs. The average increase in annual PV energy output cooled by PCMs ranged between 2% and 6% depending from the specific geographical location. A base economic analysis was also obtained and it was found that PCMs are currently not economically viable for a single-junction PVS. For a

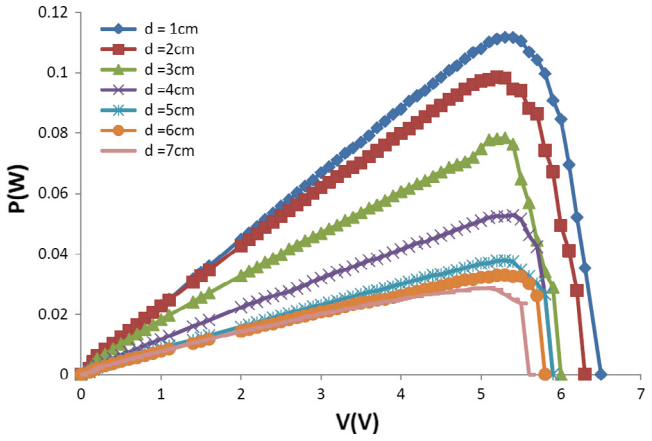


Fig. 17. Characteristic power-voltage characteristic for different water depths, [33].

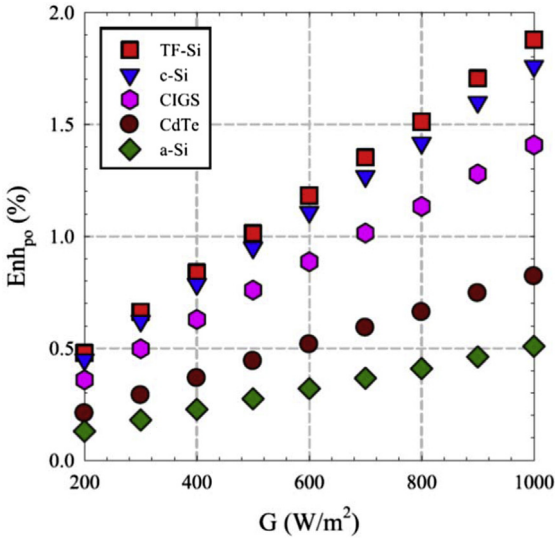


Fig. 18. Improvement in PV power output of BIPV for optimum tilt angle, [34].

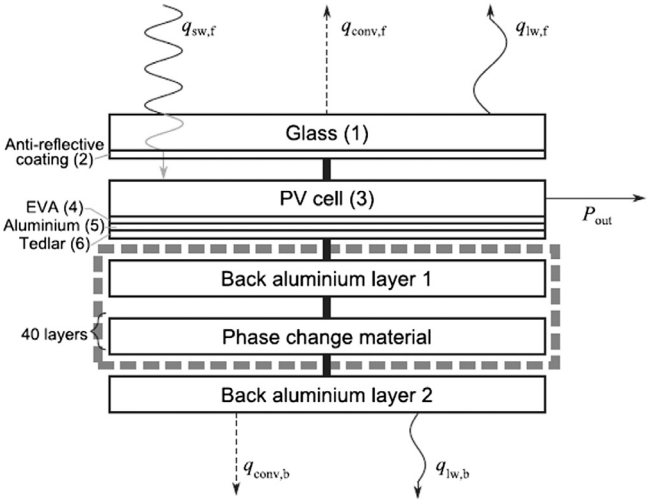


Fig. 19. Considered configurations with PCM material, [35].

0.05 thick layer of PCM material, the overall investment cost would be around €244/m^2.

According to the provided analysis, the best regions for the application of PCMs are Africa, South Asia, Australia and South and Central America, with respect to the annual increase in delivered electricity.

Radiative cooling techniques for PVs (silicon based) had been analyzed in [36] both for terrestrial and extra-terrestrial applications. A numerical analysis was obtained for temperatures that are below surrounding temperatures. The proposed radiative cooling technique consisted of a solar cell and radiate cooler. A study was also obtained for the same cooling technique in space applications (low-earth orbit). The maximal cell efficiency found for bandgaps ranged from 1.1 eV to 1.4 eV, Fig. 20, with maximal achieved efficiency ranging from 40% (terrestrial) to 45% (extra-terrestrial). A sensitivity analysis was obtained for surrounding temperatures that ranged from 300 K to 500 K and for different bandgaps (by increasing the surrounding temperature, efficiency decreases).

For an ambient temperature of 300 K, the efficiency of the cooled PV panel is higher for about 0.87% in terrestrial applications when compared to conventional PV panels with a reduction of the

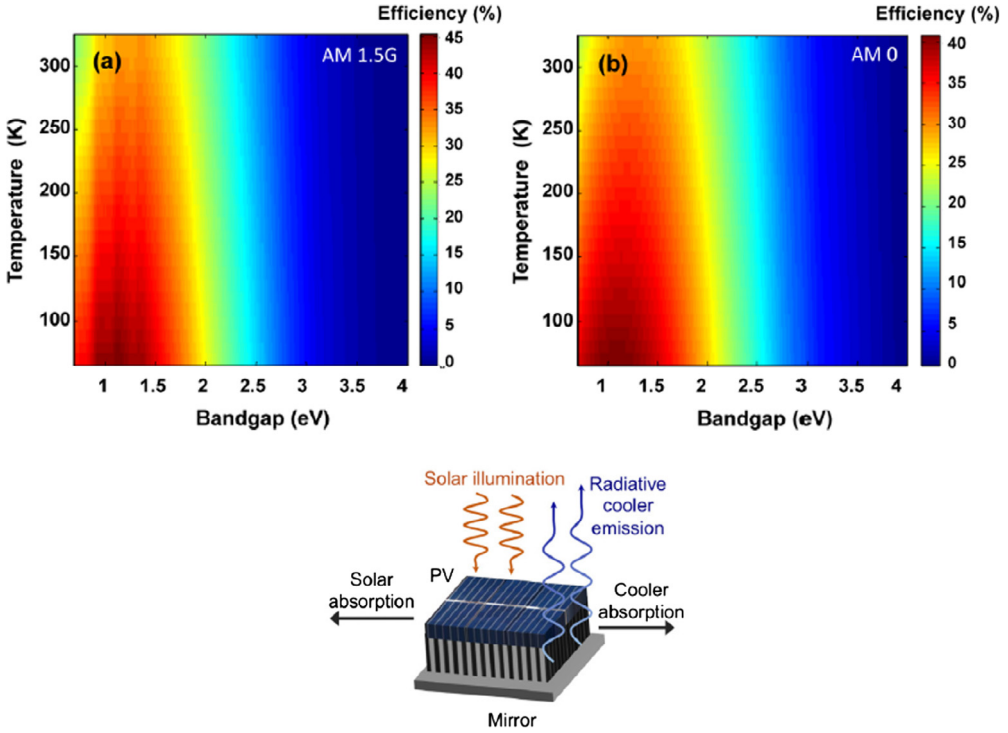


Fig. 20. Contour plot of efficiency for extra-terrestrial (left) and terrestrial (right) spectrum, [36].

operating temperature to around 35 °C (solar irradiation level of 800 W/m^2 and wind velocity of about 1 m/s). For extra-terrestrial environments, (near-earth orbit) the PV panel efficiency is higher when compared to terrestrial applications and range from 0.4% to 2.6% for the assumed operating temperatures of 293 K to 358 K. The major reason for increased PV efficiency in extra-terrestrial environments is that the radiator radiates over all the wavelengths, thus more heat radiates to the surroundings.

Paper [37] deals with PV-PCM systems where two PCM materials had been examined and addressed (PCM-$CaCl_2.6H_20$, mixture of fatty acids, capric acid-palmitic-PCM2). The PCM material was fixed in a specially designed container, Fig. 21 and mounted on the backside surface of the PV panel (installation consisted of three polycrystalline PV panels, each having a nominal power output of 85 W). The system was tested in two climate conditions, i.e. in Ireland (Dublin), with a peak solar irradiation of 990 W/m^2 with surrounding air temperatures of 16–24 °C and wind velocity from 1.0 m/s to 3.0 m/s. The second location was Pakistan (Vehari), with a peak solar irradiation of 950 W/m^2, with surrounding air temperatures of 18–34 °C, and wind velocity up to 1 m/s. It was found that the capric acid-palmitic PCM material maintained the lowest operating PV temperatures for both locations. For the Dublin climate conditions, the referent PV panel had peaked at an operating PV temperature of 49 °C while the PCM material one (PCM-2) achieved

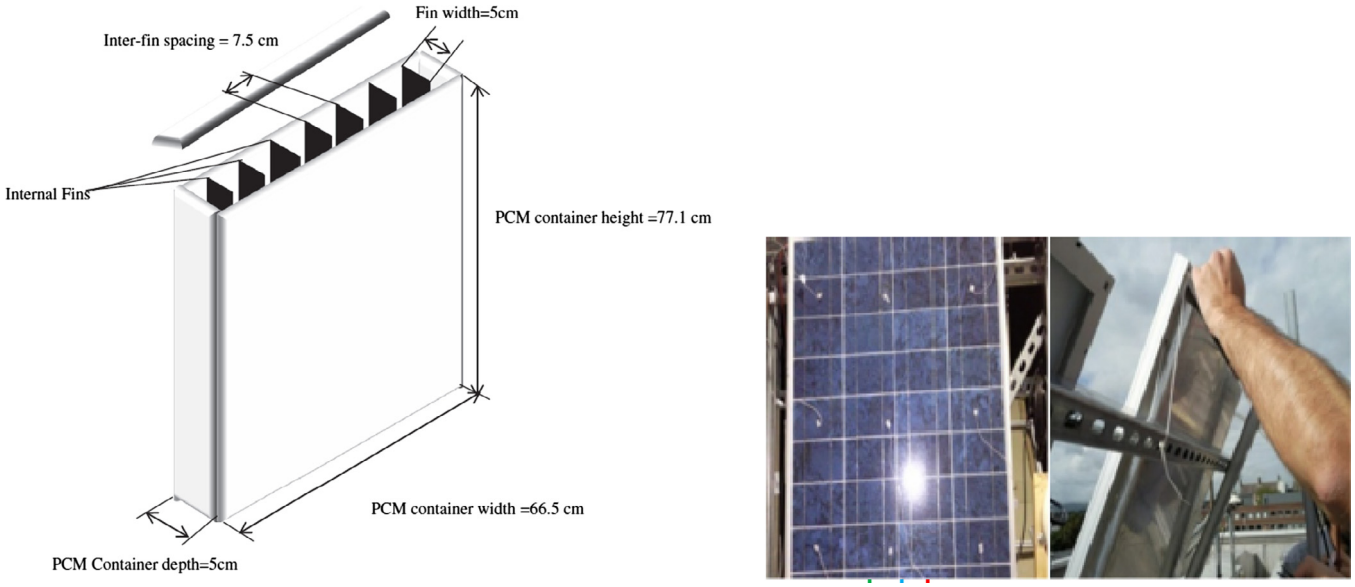


Fig. 21. Schematic of PCM container for cooling of PV, [37].

a temperature of 37 °C. For the Pakistani operating conditions, the maximal operating temperature of the referent PV panel was 63 °C, while the PCM-2 that was cooled reached about 42 °C. The average decrease in PV panel operating temperature ranged for the Ireland case from 6 °C to 10 °C and for the Pakistan case from 17 °C to 21 °C.

The maximal improvement of PV panel power output ranged from about 4% to 6.5% for Irish climate conditions (depending on the applied PCM material) and from 10% to 14% for the Pakistani climate conditions.

Paper [38] analyzed a building facility case with an integrated PCM as well as PVT systems. A simplified considered concept was presented in Fig. 22 and numerically analyzed for Australian climate conditions (results were validated with available experimental data). Three types of PCM materials had been considered (RT18HC, SP21E, SP24E) with a variated thickness of 5–30 mm and with melting temperatures ranging from 19 °C to 25 °C (the air flow was variated from 1000 kg/h to 2000 kg/h). The main idea of the research was to provide optimization for the buildings with related PCM materials where the Taguchi method was used together with an analysis of variance (ANOVA). Three different scenarios had been analyzed, the first one being a house with PVT collectors only, the second house with PVT collectors and an integrated PCM, and the last one was a house that only applied a PCM material. Optimal thermal performance improvement was found for a 200 mm thickness of the mentioned PCM materials with an increase from 43% to 46%. However, the authors did not address the PV part of the system in this paper so the data related to the general PV characteristics as well as performance data were not specified.

An analysis of BIPV systems with integrated PCMs was provided in [39] with obtained LCA analysis, Fig. 23. The main focus of the paper was to elaborate the operational control of the BIPV systems for different climates, for cases with and without the application of PCM materials (hydrate salt, i.e. Glauber salt-PCM1, $Na_2HPO_4{\cdot}12H_2O$-PCM2, and 50%PCM1 + PCM2 mix). An overview of existing BIPV concepts was also provided with special emphasis on the type of PCM material used. For the purpose of mathematical modelling, a polycrystalline PV system was considered with a nominal power output of 250 W (and rated conversion efficiency of 11.7%). Numerical simulations were done for a solar irradiation level of 800 W/m^2, with a surrounding air temperature of 20 °C and wind velocity of about 1 m/s (Spain - Zaragoza climate conditions were considered with an inclination angle of 35°). The BIPV system without a PCM material reached a maximal temperature of 48.8 °C with a related electrical efficiency of 10.58%. By applying a PCM material, the temperature was reduced from 3 °C (Glauber salt) to 10 °C (PCM mix) during the periods of highest solar irradiation levels and depending from the type of PCM material. From the conducted environmental analysis, the authors concluded that

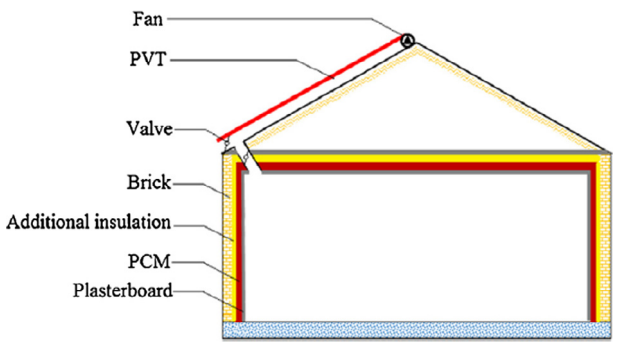


Fig. 22. Concept of house with integrated PCM as well as PVT ventilation system, [38].

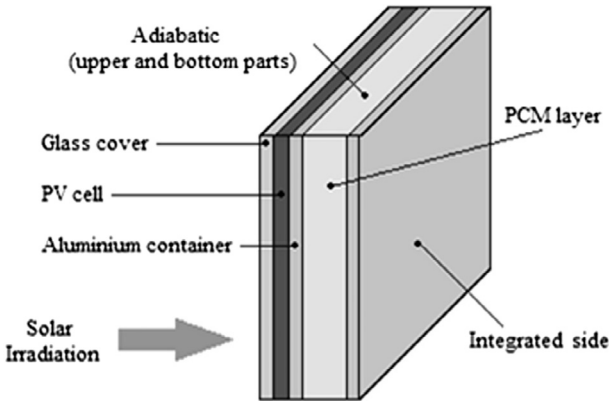


Fig. 23. Configuration on which analysis in [39] was based.

the application of PCM materials reduces the premature degradation of PV panels with an extended PV lifetime ranging from 28 years to 48 years. Finally, it was concluded that BIPV systems with PCM materials are prospective from a technical and environmental point of view, and that additional efforts need to be provided for intense market penetration.

Paper [40] dealt with experimental work where a building integrated concentrated photovoltaic system was investigated (BICPV). Besides the examination of thermal regulation issues through application of PCM materials, authors also examined the electric response of the proposed concept, Fig. 24. The considered system was equipped with a linear asymmetric compound parabolic concentrator and was tested under laboratory conditions for solar irradiation levels ranging from 500 W/m^2 to 1200 W/m^2. The used PCM material was the RT-42 and the PV cells were silicon based. Absolute increase in electrical efficiency ranged from 6.48% to

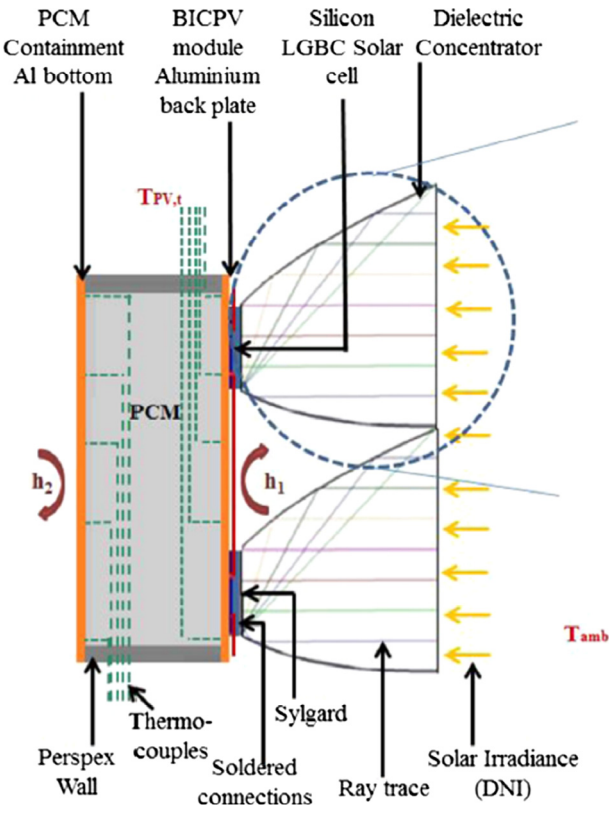


Fig. 24. Experimental setup for proposed BICPV solution, [40].

6.98%, for a solar irradiation level of 1000 W/m^2. The relative increase in electrical efficiency ranged from 1.15% up to 6.8%, depending from the solar irradiation level applied. The maximal PV panel operating temperature was 55.2 °C without PCM materials and 49.2 °C with PCM materials. The maximal reduction of the PV panel operating temperature was about 5 °C and averaged at around 3.8 °C (the highest reduction was for a solar irradiation level of 750 W/m^2). The maximal achieved increase in PV panel output was 5.2%. Therefore, the main contribution to the considered research study was the investigation of electrical response besides temperature and melt fraction issues.

A selective-spectral as well as radiative cooling of photovoltaics was considered in [41]. The authors numerically analyzed sub band-gap absorption and thermal radiation for different PV technologies (Si, CdTe, CIGS, GaAs). The study resulted in them recommending a redesign of the optical properties for PV panels in order to enhance radiative cooling. The main idea was to eliminate energy absorption in order to enable selective spectral cooling. According to the authors, the proposed cooling technique would reduce the PV panel operating temperature by 10–20 °C (with an increased lifetime of the PV panel as well as a reduced LCOE). The optic and electrical properties of the PV modules had been addressed and the general cause for PV panel self-heating (sub background absorption). Furthermore, different radiative-based cooling techniques had also been elaborated such as the selective spectral cooling method and general radiative cooling method in order to reduce PV module self-heating. A comparison of the selective-spectral and radiative cooling techniques was also done, Fig. 25. According to the authors, the expected absolute increase in efficiency could be from 0.8% to 1.8% and the extended PV panel lifetime could range from 80% to even 260% for Si solar modules. Finally, the authors stressed that the selective-spectral cooling technique is more favorable than the radiative cooling technique when taking different aspects into account that are addressed in detail in the paper.

The authors in [42] proposed an active and passive cooling technique for photovoltaic panels using forced air flow induced by fans at the backside surface of the PV panel, Fig. 26, as well as a natural convection case. Different widths of free space below the PV panel were examined in the case of free convection (from 0.105 m to 0.165 m) as well as different fan-induced air velocities over the PV panel (ranging from 2 m/s to 4 m/s). The panels were tested for climate conditions in Spain (Cartagena) with a solar irradiation peak of 1000 W/m^2. In general, different polycrystalline PV panels were examined with the nominal power output ranging from 240 W to 280 W. In the case of natural convection, two PV panels were used, i.e. a referent one and the second one was placed above a steel plate with an air channel underneath (with variable space as already mentioned). The results for the PV panels mounted on the roof of the industrial building showed that in the case of natural convection, the second PV panel was warmer (with maximal improvement of electrical efficiency being at about 0.5%). The average temperature difference between the examined PV panels was 6 °C, however, when the space between the PV panel and roof was the smallest, the temperature difference ranged from 8 °C to 10 °C (at high irradiance, i.e. at 970 W/m^2). The authors emphasized the importance of distance between the PV panel and roof and generally concluded that larger aspect ratios are desirable in order to increase heat rejection from the PV panel for both the passive and active examined cooling techniques.

In study [43], an experimental approach was obtained for both passive and active PCM based cooling techniques as presented in Fig. 27. The first panel in Fig. 27 was with a PCM (RT42), the second one was referent and the third one was a PV panel with a PCM (about 10.2 kg) which was additionally cooled with water. The highest obtained temperature decrease in the PV panel operating temperature was about 19 °C and the best effect was obtained in the PV-PCM-water system. However, there is not a significant difference in operating temperature reduction when PV-PCM and PV-PCM-water are compared (the PV-PCM-water system is slightly better). The average temperature reduction was about 5 °C between the conventional PV panel and the one with the PCM and the average temperature drop for the PV-PCM-water system was about 10 °C. The average increase in PV power output was 6% and the increase in PV electrical efficiency was about 1%.

3. Elaboration of considered passive cooling techniques and general comparison

Based on the previously considered and elaborated relevant references, a summary of the main influential parameters as well as important general information was presented in Table 1, for each considered passive cooling technique. A synoptically presentation was obtained in order to analyze current progress, as well as to provide base data for upcoming economic evaluation. The data that are presented in Table 1 are specified either as a single number (which are in that case peak values) or as a value range; in both

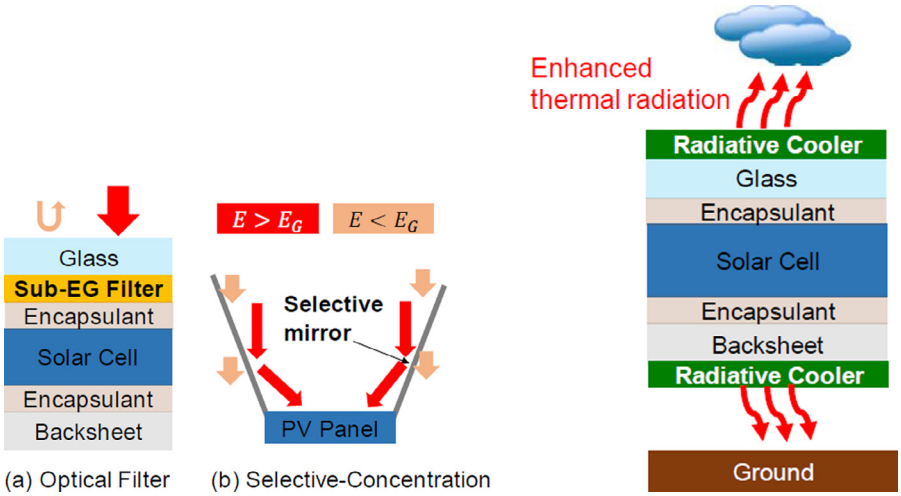


Fig. 25. Selective spectral cooling (left) and radiative cooling (right), [41].



Fig. 26. Experimental setup for air based cooling technique (left-natural convection configuration and right forced convection circulation), [42].

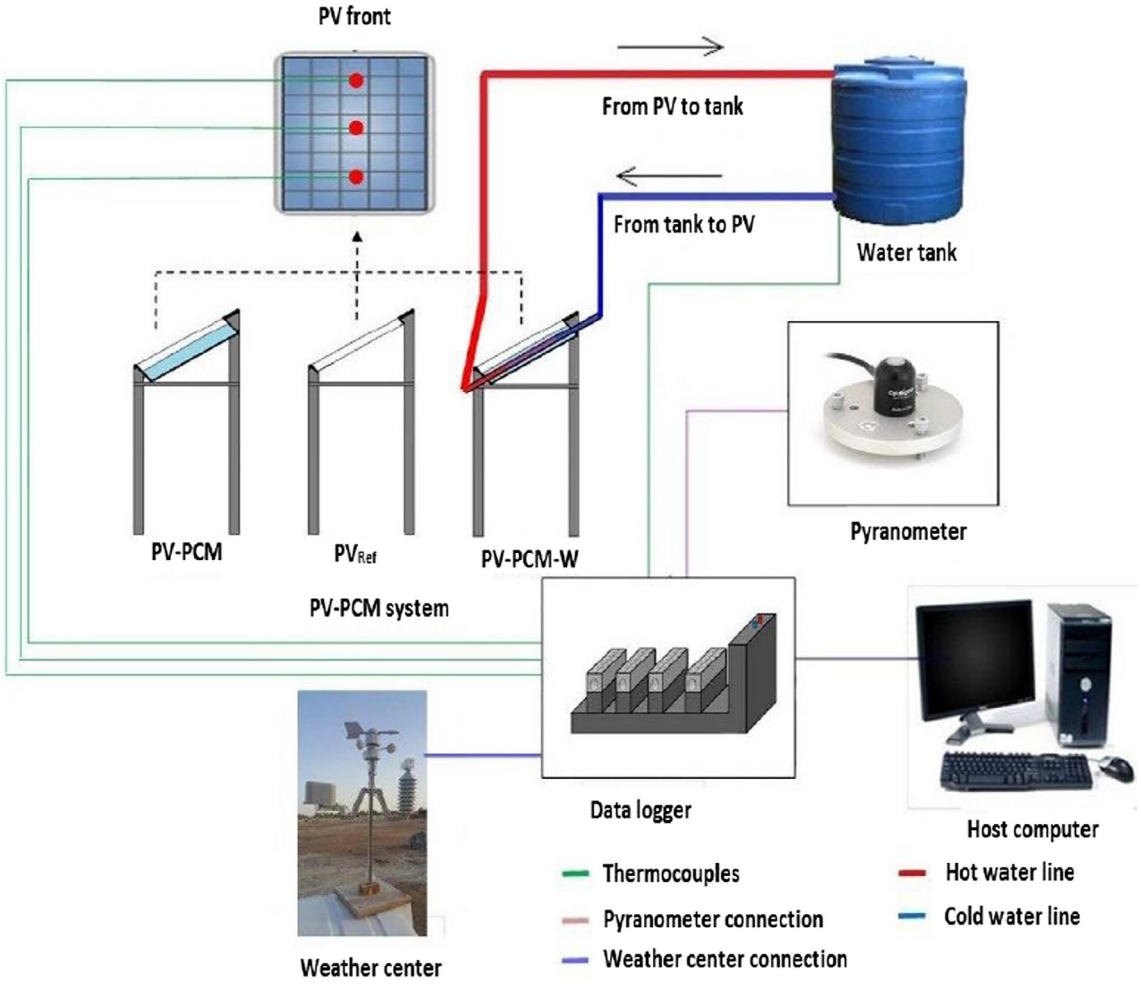


Fig. 27. Passive and active PCM based cooling technique for PVs, [43].

cases the numbers were presented based on the general availability of the reported data for the herein considered references. Further, the underlined values in Table 1 are calculated and based on available data for temperature reduction related to the PV operating temperature (average degradation for Si-based PV technologies was assumed to be about 0.4%/°C, which was a necessity in order to be able to obtain sufficient economic evaluation for the herein obtained approach where general economic aspects were examined). Certain data in Table 1 that were not reported by the authors in the analyzed research studies were labeled as "N/A".

Besides the previously specified PV panel performance characteristics, general PV technology types, coolant types

Table 1
Overview of general data as well as influential parameters for considered passive cooling techniques for PVs.

Ref.	*PV* *Technology*	*Coolant/ technique*	*Efficiency* *Increase*	*Increase in the panel* *Power output*	*Reduction in the panel* *Operating temperature*	*Economic* *Aspect*	*Location/solar* *Irradiation level/suns*	*Other general* *Circumstances*
[17]	Si	Air	15.2%	50%	38 °C	N/A	Victoria (Australia)	Field exp.
[18]	Si	PCM	12%	N/A	30 °C	N/A	Newtownabbey (UK)/750 W/m^2	Field exp.
[19]	Si	Air	4–8%	18.3%	10–20 °C	N/A	500–1000 W/m^2	Numerical exp.
[20]	Si	Water/ immersion	16–20%	52.6%	40–50 °C	N/A	Pisa (Italy)/600–800 W/m^2	Field exp.
[21]	Si	PCM	4-7%	N/A	10–18 °C	N/A	500–1000 W/m^2	Laboratory exp.
[22]	Si-mono	Air	20%	24%	6.6–17 °C	N/A	200–800 W/m^2	Laboratory exp.
[23]	Si	PCM	8.4%	N/A	21 °C	N/A	750 W/m^2	Laboratory exp.
[24]	Si	PCM	12%	N/A	30 °C	N/A	200–600 W/m^2	Numerical exp.
[25]	Si (10 W)	PCM	9.4%	55%	15 °C	N/A	600–2300 W/m^2	Field and lab. exp.
[26]	Si	Air	3.2%	N/A	8 °C	N/A	1000 W/m^2	Numerical exp.
[27]	Single crys. Si (50 W)	Nanofluids	9.4–15%	8.8–15%	9–20 °C	N/A	Tiruchirappalli, (India)/ 0.1–0.5 MJ/ m^2	Field exp.
[28]	Mono Si	Liquids/ immersion	8.5–15.2%	N/A	N/A	N/A	10–30 suns	Laboratory exp.
[29]	Si	PCM	20%	N/A	50 °C	N/A	1,000 W/m^2	Laboratory/ numerical
[30]	Mono Si (10 W)	PCM	6–21.2%	N/A	0.4–4.4 °C	N/A	Indonesia (Bandung)/1049–1120 W/ m^2	Field exp.
[31]	Poly Si (10 W)	Water/ evaporation	16%	14.2–19.1%	40–50 °C	N/A	United Arab Emirates (Sharjah)/ 900 W/m^2	Field exp.
[33]	Poly Si (0.119 W)	Water/ immersion	11–22%	N/A	N/A	N/A	Iraq (Dohuk)/700 W/m^2	Field exp.
[34]	Various technology	PCM	4%	N/A	10 °C	N/A	200–1000 W/m^2	Numerical exp.
[35]	Si	PCM	2–6%	10%	N/A	Yes	Different world regions/350 W/m^2	Numerical exp.
[36]	Si	Air/radiative	0.4–2.6%	N/A	35 °C	N/A	800 W/m^2	Numerical exp.
[37]	Poly Si (65 W)	PCM	2.4–8.4%	4–13%	6–21 °C	N/A	Ireland (Dublin), Pakistan (Vehari)/400 W/m^2, 900 W/m^2	Field exp.
[39]	Poly Si (250 W)	PCM	3%	N/A	3–10 °C	N/A-LCA	Spain (Zaragoza)/800 W/m^2	Field exp.
[40]	Si	PCM	1.15–6.8%	5.2%	3.8–5.0 °C	N/A	500–1200 W/m^2	Laboratory exp.
[41]	Various technology	Air/radiative	0.5–1.8%	N/A	10–20 °C	N/A	100–1000 W/m^2	Numerical exp.
[42]	Poly Si (240–260 W)	Air	4.8%	7.5%	6 °C	N/A	1000 W/m2	Field exp.
[43]	Poly Si (40 W)	PCM	2.5–5%	6–9%	5–10 °C	N/A	Al Ain (UAE)/970 W/m^2	Field exp.

(i.e. techniques), economic aspects, specific geographical location characteristics, and finally other general circumstances are also summarized in Table 1.

The first general notice is that more than half of the examined research studies are related to the application of PCM materials as a means of cooling and in general for controlling PV thermal performance. Other papers consider either liquid based cooling options (water or nanofluids) or air based cooling techniques and, finally, the less investigated radiative based cooling techniques. Furthermore, an economic aspect was considered in only one research study, namely in paper [29] which provided an LCA for the PCM based cooling method). It is also interesting to notice that there is approximately the same number of laboratory, numerical and field proposed cooling techniques for PVs that were examined. The majority of the PV technologies examined was Si-based and usually exposed for a peak average solar irradiation level that ranged from about 800 W/m^2 to 1000 W/m^2.

According to the analysis of the data summarized in Table 1, the main performance parameters, i.e. their value range was specified in Table 2, and classified according to the considered cooling technique.

In Table 2, it is noticeable that PCMs, liquid and air based cooling techniques for PVs can achieve a maximal electrical efficiency increase of about 20%, whereas the highest recorded increase of 22% was obtained for the water immersion method, enabling a reduction in the PV panel's operating temperature which was up to 50 °C. The lowest found improvement in efficiency was for the radiative based cooling technique, a little less than 3%. A minimal improvement in efficiency, as well as a PV operating temperature reduction was found for the water based cooling technique. Thus, liquid and air based cooling techniques as well as PCMs have a certain potential for the improvement of efficiency from a pure performance point of view, although improvements are needed for it to be achieved in an economically viable way. Finally, other aspects also need to be taken into account, besides panel performance, when considering certain cooling techniques for PVs. It is also important to stress that the maximal values in Table 2 are basically peak values and average values. For PCMs and air based cooling techniques, the average increase in delivered electricity yield would certainly be less than 10%, the average for liquid based

Table 2
Summary of main performance parameters for herein considered passive cooling techniques for PVs.

Cooling technique	*Efficiency Increase*	*Increase in the panel Power output*	*Reduction in the panel Operating temperature*
PCM	1.15–21.2%	4–55%	0.4–50 °C
Liquids	8.5–22%	8.8–52.6%	9–50 °C
Air	3.2–20%	7.5–50%	6–38 °C
Radiative	0.4–2.6%	Non	6–21 °C

cooling techniques would be between 10% to 15% and about 1% to 1.5% for radiative.

4. General economic evaluation: 30 kW PV system case study for operation in Mediterranean climate conditions

To check the economic feasibility of different cooling techniques, a case study related to a 30 kW photovoltaic plant that operates in typical Mediterranean climate conditions was considered, City of Split (Croatia), Fig. 28 (general data related to the considered PV plant are presented in Table 3). The PV system was constantly monitored via SUNNY PORTAL web interface, thus real data are available for the produced electricity yield, i.e. PV system performance. For the period of 2014–2016, the average annual electricity production was about 42,000 kW h/year and the average estimated annual O&M cost is about 660 €/year (the PV system has been in operation for three years). The payback time of the PV system is around 4 years, with included governmental subsidies that are approximately in the amount of 13,500 €/year. Without governmental subsidies, the payback time period would be doubled, i.e. varying between 8 and 9 years.

Until March 2017, the PV system has made revenues of approximately 41,000 € and helped in avoiding emissions of about 109,000 $kgCO_2$, it is thus expected that the system will be paid off by the already mentioned payback time period of 4 years. However, an economic analysis will be obtained for the case without governmental subsidies within the frame of this paper, as this is the trend for all European countries and for the reason that PVs are increasing in competitiveness [44].

If the general acceptable LCOE calculation framework is applied, then the LCOE can be calculated as follows, [45],

$$\mathrm{LCOE} = \frac{\mathrm{AC}}{\mathrm{EO}} = \frac{\mathrm{IC} \times \mathrm{CRF} + \mathrm{OM}}{\mathrm{EO}}, \quad (1)$$

where the capital recovery factor (CRF) is derived from,

$$\mathrm{CRF} = \frac{(1+p)^n \cdot p}{(1+p)^n - 1}. \quad (2)$$

If interest rate is taken at about 6% with an amortization period of 30 years (usually values considered for PV systems), the calculated LCOE for the considered 30 kW PV system equals 0.102 €/kW h. This result is in accordance with LCOE related projections and obtained by Fraunhofer ISE, [46]. Namely, LCOE data in [46] are specified for current market available PV technologies where LCOE ranges from 0.08 €/kW h to 0.14 €/kW h.



Fig. 28. Considered 30 kW PV system.

Table 3
General data related to 30 kW PV system.

General data for the 30 kW PV system	
Photovoltaic technology	Si-polycrystalline
Overall nominal power output	30 kW
Nominal panel power output	250 W
Overall PV panel surface	204.6 m^2
Nominal efficiency	15.37%
Number of PV panels	126
Overall investment	49,700 €
Average produced electricity	42,070 kW h/year
Annual O&M cost	660 €/year

To check the economic viability of the passive cooling techniques, available data had been taken from the literature, although it has to be underlined that there is a lack of data related to the expected overall investment for the examined cooling techniques, as well as a lack of certain data on technique performance. Due to those reasons, reasonable assumptions for overall investment were made, when no detailed data were available, so as to be able to estimate the LCOE range and assess the economic feasibility as a result. Finally, an analysis was obtained regarding the passive cooling techniques for the considered PV system and its assumed coolant type (PCM, liquid or air as the three most viable cooling techniques). Radiative cooling techniques were not considered in the economic analysis due to their rather poor performance, at the present stage of this research. Therefore, the main idea was to evaluate the technical possibility as well as economic aspect related to the application of specific passive cooling options for a realistic PV system.

4.1. PCM based cooling technique

The first considered cooling technique is the application of a PCM, as it was considered in the majority of research reports. Authors in [35] reported the overall investment for PCM based cooling to be at about 244 €/m^2, with an assumed 5 cm thick PCM layer and an increased average electricity yield of about 3–7%, depending on the specific location. This figure is a realistic expectation which is also based on other existing field studies. Therefore, the overall investment to merely adapt our existing PV system to one that would be cooled with a PCM would be about 99,600 €. In the previously specified overall cost, we did not include the cost of aluminum cast and which according to the data provided in [35] is expected to be about 5300 € (1.60 €/kg and about 3315 kg of aluminum). Furthermore, it is reasonable to expect that the cooled PV system would increase the average delivered electricity by about 3–6% for the considered location of the PV system, [35], i.e. from about 43,332 kW h/year to 44,595 kW h/year. It is also reasonable to expect a prolongation in overall lifetime for the PV plant by at least 30–40 years approximately, as cooling techniques generally have favorable effects and long term reductions in electrical efficiency degradation. Namely, the authors in [39] reported that by applying PCM cooling techniques, PV panel lifetime could be prolonged up to 48 years; an LCA analysis was obtained in the same study. Finally, LCOE was calculated for the considered 30 kW plant, with the same economic input parameters as in the case of non-cooled PV systems, except for the overall lifetime period taken which was 40 years. Therefore, taking PCM cooling, the LCOE will range from 0.171 €/kW h to 0.176 €/kW h, depending on the expected increase in electricity yield, namely from 3% to 6%. It is important to emphasize that the O&M cost had been considered in the calculations as the same, although it is reasonable to expect an increase due to the PV/PCM configuration. Meanwhile, it is clear that PCM based cooling for the assumed

PV system is not economically viable. Even when taking a hypothetical case of 21% increase in electricity yield, which is the highest value mentioned in the literature for PV/PCM systems, it can certainly not be an average one, and when also considering a maximum reported expected lifetime of 48 years, the LCOE would still be at about 0.15 €/kW h, with all other input parameters remaining the same. Even in the previous hypothetical case, the LCOE numbers are still not economically competitive, when comparing cooled PV/PCMs to non-cooled PV systems.

Thus, the main issue is to determine the upper limit of the overall investment for the PV/PCM system, which would still make it a feasible investment. If we assume a targeted increase of about 10% in PV/PCM system performance, and an extension in the system's lifetime of about 40 years, we can project the allowed overall cost for the considered PV system expressed in €/m^2, with the all other parameters remaining the same. For those circumstances, the maximal allowed investment cost to obtain minimal economic benefit and a reasonable LCOE is about 6000 €. Thus, a maximum incremental cost of around 30 €/m^2 would be allowed for the application of the PCM cooling technique, which would result in a LCOE of 0.098 €/kW h. In Ref. [35], the typical reported price average of the PCM was about 4.93 €/kg. Hence, for the considered case of a 0.05 m thick PCM layer, the additional required mass would be about 43 kg/m^2. Taking into account the specified values in [35] for the herein considered 30 kW PV system, the overall cost would be about 344 €/m^2 when only talking about PCM, not to mention the additional aluminum cast with a cost of about 32 €/m^2, [35]. It is also important to stress that the overall additional roof weight for the considered cooling system is expected to be about 59 kg/m^2. The existing installed PV system is about 35 kg/m^2, the overall additional roof weight would thus be about 94 kg/m^2 and which is tolerable but is upper limit regarding static. The latter is at approximately 120 kg/m^2, which is the maximal allowed additional weight for the existing concrete roof where the PV system is currently mounted. It is therefore evident that a significant reduction in PCM price is needed for this cooling technique to become feasible, as well as novel design solutions in order to have more efficient PV/PCM systems overall, which would also allow easier installation and maintenance.

4.2. Air based passive cooling technique

According to the review provided, passive air based cooling techniques for PVs can increase the delivered electricity from about 3% to peak values of around 20%. The reported maximal efficiency peak of 20% in [22] was obtained for small single sized PV cells cooled with fins, which can hardly be the case for standard sized PV panels. Namely, for standard systems that consist of a few hundred PV panels, the situation would be rather different and it could be reasonable to expect an average increase in delivered electricity of about 3–5%. This has to do with the heat rejection from a great number of panels, that reduces the potential for passive cooling in each PV element alone, as the heat rejection of one panel affects the air flow around its neighboring ones, (the previous is in accordance with the examined existing field studies and it is a realistic expectation, [19,26]). The previously specified average increase in efficiency could be ensured for the case of simple PV panel cooling with aluminum fins. Therefore, a passive cooled PV plant with aluminum fins had been taken for the herein considered case (about 1 mm thick) that are assumed to be mounted on the backside surface of the PV panel. The fins can be retrofitted by means of fixing them with epoxy glue containing additives to enhance thermal conductivity. The average mass consumption of the 1.0 mm thick aluminum is about 0.9 kg/m^2 and of around 0.1 kg/m^2 for the epoxy glue (the epoxy glue lifetime is more than 50 years if not exposed to direct sunlight) for this case, the previously specified numbers are reported in correspondence with the company [47] and by existing experimental experience [48]. For the herein considered 30 kW PV system, the estimated cost for aluminum fins would be about 408 € (2.20 €/kg with labor included to prepare Al fins) and about 1044 € for epoxy glue containing additives (approximately 5.1 €/m^2 including labor cost for the preparation of epoxy glue containing additives). The overall expected cost for labor would be about 760 € (approximately two panels per hour and labor cost of about 12–15 €/h).

Finally, the overall expected investment for the considered passive cooling option and 30 kW PV plant was estimated to be at about 2200 € (i.e. around 10.8 €/m^2). If the previously specified data are taken for LCOE calculation, with all other input parameters remaining the same as in Section 4.1, the LCOE ranges from about 0.093 €/kW h to 0.095 €/kW h. Thus, from the previously gained numbers it is clear that the considered passive air based cooling technique for PVs is an economically viable option for the 30 kW PV system case.

4.3. Liquid based passive cooling technique

The liquid based cooling technique for the considered PV system is primarily based on the liquid immersion method where PV panels are assumed to be flooded in a certain liquid. According to the previously obtained comprehensive review of passive cooling options for PVs, water turned out to be the most suitable and reasonable coolant for the considered PV system, whereas other liquids may be referable to CPV systems.

The authors in [43] based their reports on experimental data, maximal increase in conversion efficiency being at about 20% for the flooded PV panel case with a 6 cm thick water layer. Further, for typical Mediterranean climate conditions (Italy, Pisa), the authors in [20] also tested the performance of the water flooded PV panels and reported an average increase of about 11% in PV panel efficiency for a 4 cm thick water layer (the average water temperature remained at about 30 °C). Therefore, it is reasonable to expect the average increase in efficiency for the flooded PV system with a 6 cm thick water layer to roughly range at about 10% up to 15%. It is also important to mention that the flooded PV panels would be positioned in a horizontal position, which means that they would not be able to deliver the same yearly quantity average of produced electricity when compared to the existing PV system that is set to an optimal angle of about 27°. Thus, the delivered electricity reduction of the considered PV system would certainly be at about 15%, i.e. meaning around 35,800 kW h/year of annual delivered electricity (PV system data from Table 3). Therefore, the overall annual delivered electricity would range between 39,380 kW h/year and 41,170 kW h/year (average at about 40,300 kW h/year).

The 6 cm thick water layer would cause about 60 kg/m^2 of additional weight on the roof, i.e. for PV modules included; the overall weight would be about 95 kg/m^2. Further, the basin weight as itself should be added to the previously estimated additional roof weight, (i.e. a few basins most likely due to technical limitations and general issues related to civil engineering aspects, i.e. for the herein circumstances, three basins should be sufficient for the considered number of PV panels). The basins can be made of concrete, polypropylene or even fiberglass as most realistic options.

For the existing situation, where it is assumed to mount basins on the existing roof, the acceptable roof weight would be an issue. Namely, about 18 m^3 of concrete would be needed for the concrete option and which means around 270 kg/m^2 of additional roof weight, which is above the allowed static limit for the existing construction, even without considering the weight of the water layer and the PV construction in general. Thus, concrete basins could only be an option in the case of new buildings, where static

construction reinforcements are ensured. The cost of concrete basins could also be an issue from an economic point of view as the average price of reinforced concrete with the included labor is about 150–200 €/m^3 and the average sealing cost is about 20 €/m^2. Finally, the overall expected investment is roughly about 7000–8000 € for the concrete option, which leads to an LCOE of about 0.11 €/kW h and thus leading to concrete not being an option at all for our case (we namely have the same remained level of O&M cost like in Table 3, however, it is more realistic to expect a higher annual O&M cost, which will additionally increase the LCOE above 0.11 €/kW h in the end. Therefore, polypropylene materials (or even fiberglass) could be a more realistic option, where the cost of three polypropylene basins would roughly be about 40,000 €. The overall additional roof weight in that case would be at an upper limit of about 110 kg (60 kg/m^2 of water, 35 kg/m^2 of PV panels and about 15 kg/m^{2}of polypropylene basins). Based on the previous elaboration, it is clear that the LCOE value is not competitive for even polypropylene basin cases: an LCOE of about 0.185 €/kW h was calculated, for all other parameters it is as in Section 4.2., although the O&M would definitely be higher due to the fact that the basins need maintenance. The main economic point is proven anyway, together with the complexities introduced due to increased roof weight.

Hence, taking into account the detailed elaboration presented in this section, with respect to the studied case of a 30 kW PV system, the only currently economically viable and technically reasonable solution is the passive cooling option, with the use of aluminum fins and the surrounding air acting as a coolant; the figures are synopsized in Table 4.

The other possible cooling options cannot be considered as economically feasible and viable, mainly due to their high initial investment. Therefore, although PCMs and liquid cooling options can ensure reasonable increases in the performance of PV systems, they still cannot compensate the relatively high investment which leads to a non-favorable LCOE.

5. Environmental aspect of considered passive cooling techniques

The paper also targets on evaluating the environmental impact of the above considered cooling techniques by using the LCA approach, [49]. A life cycle assessment had been performed following the international standards of ISO 14040 series, [50–53], supported by Gabi Software, [54]. The aim is to trace the system's environmental impact related to its life cycle. Life cycle thinking in environmental impact assessment has been present for over 40 years now, predominantly in the industrial and service sector, with some early Life Cycle Analysis (LCA) type practitioners bringing a multi-attribute quantitative approach to decisions related to beverage packaging as far back as the 1960s. This approach evolved into LCA, with well-established international standards for guidance. LCA is therefore a science-based, quantitative and integrative methodology that measures materials and energy flows to and from nature over the lifetime of a product or service, and assesses the potential impact of those flows on resources, ecosystems and human health (carbon footprint, acidification, eutrophication, photochemical oxidant creation and ozone depletion).

Table 4
General LCOE data related to 30 kW PV system and considered passive cooling options.

General LCOE (€/kW h) data for 30 kW PV system	
Referent PV system	0.102
PCM based cooling	0.171–0.176
Air based cooling (Al-fins)	0.093–0.095
Liquid based cooling	0.11–0.185

The main purposes of LCA are to identify hot spots in the value chain, identify and quantify alternatives, and disclose environmental information. The following phases have been evaluated: production and transportation of energy and raw materials, production process, installation, maintenance, disposal and transports occurring during each step. The LCA approach used is best termed as "streamlined" LCA (SLCA), as it does not take into account recycling or disposal into landfill.

LCA therefore offers a comprehensive analysis which links actions with environmental impacts. At the same time it provides quantitative and qualitative results and takes into consideration the link between system functions and environmental impacts and it is easy to identify the issues that need improvement. There are four main stages for the implementation of LCA: (a) planning, (b) inventory analysis, (c) impact assessment and (d) improvement analysis. The main purpose of planning is to define the scope and goal as well as the boundaries of the system studied. During the planning, the objective should be clear and the data collection sources defined. The inventory analysis is the stage where the inputs and outputs are quantified. Energy, raw materials, water consumption, air emissions, solid waste, are quantified either through measurements or database searches, surveys and software calculations.

During impact assessment, the environmental impact is calculated as a result from the inventory analysis. The input and output data are translated to environmental impacts like climate change, acidification, eutrophication, and photochemical oxidation. Finally, the stage of improvement analysis involves discussion and improvement suggestions. For implementing LCA at the inventory phase, a software tool was used for the results' reliability control: the GABI LCA software, which is a life-cycle analysis model with embodied EcoInvent LCA. At the environmental impact assessment phase (normalization and weighting), a set of specialized indicators were used, derived from CML 2.

CML 2001 is an impact assessment method which restricts quantitative modelling to early stages in the cause-effect chain to limit uncertainties. Results are grouped in midpoint categories according to common mechanisms (e.g. climate change) or commonly accepted groupings (e.g. ecotoxicity). CML 2001 was developed by the Institute of Environmental Sciences, Leiden University, The Netherlands, and was published in a handbook with several different authors, [55]. The main principles behind the methodology are not being further developed. A Microsoft Excel spreadsheet with characterization factors for more than 1700 different flows can be downloaded from the CML website. The characterization factors are updated when new knowledge on the substance level is available.

Several additional characterization factors are calculated by PE and LBP-GaBi following the principles described in the CML 2001 methodology documents. Normalization factors for CML 2001 are available for the Netherlands, Western Europe, EU and the World. The normalization factors are calculated via total substance emissions and characterization factors per substance, and hence follow the substance level updates as described above.

Normalization data for other countries (e.g. Germany, USA and Japan) and geographical units (e.g. North America or OECD) is available in GaBi5.

The four passive cooling techniques compared and analyzed by means of technical characteristics, economic effectiveness and environmental impact are in brief:

(1) The referent PV system of 30 kW with an overall surface of 204.6 m^2 and which is actually the referent point for comparison.

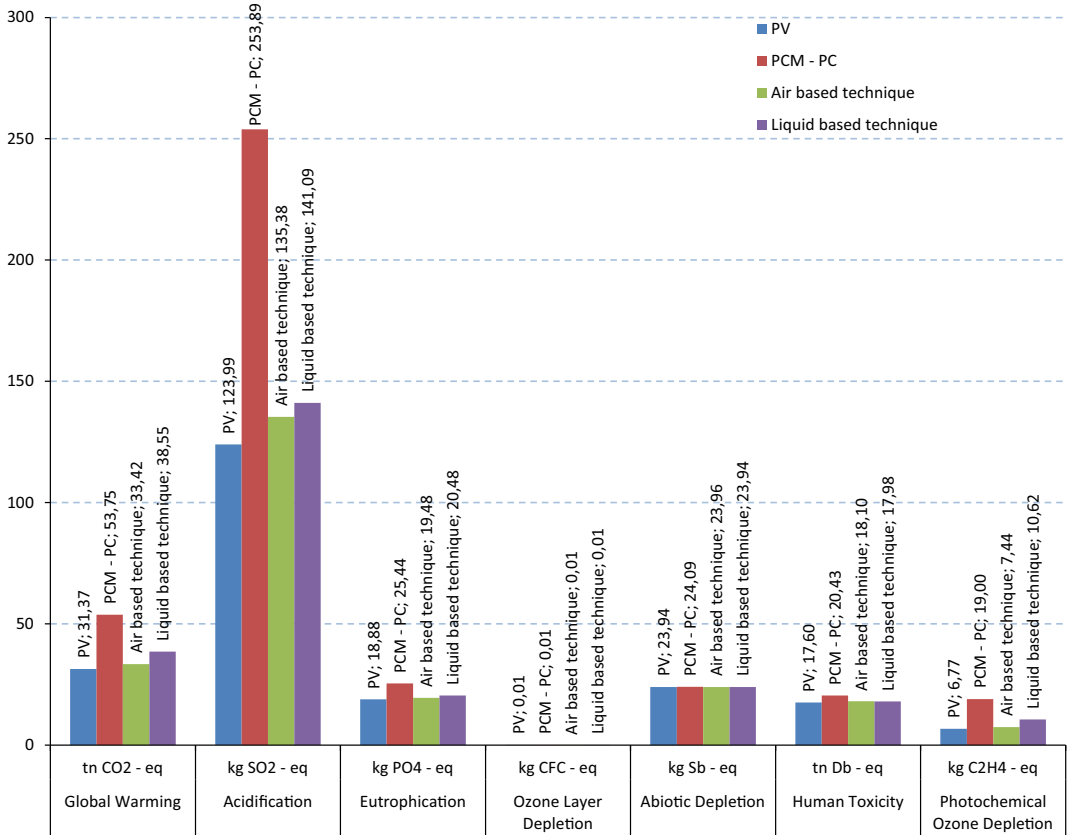


Fig. 29. Environmental performance of different cooling techniques based on LCA methodology.

(2) PCM based cooling technique.
(3) Air based passive cooling technique.
(4) Liquid based cooling technique.

According to the obtained LCY analysis, it is obvious from Fig. 29 that the PCM cooling technique is the one with the most significant environmental impact, especially when considering global warming and the acidification environmental impact. This is caused because the PCM suggested cooling technique consists of aluminum and glass components as well as paraffin based materials (therefore compared to the other cooling techniques, PCM choice causes the most significant environmental impact), Table 5. Nevertheless, considering abiotic depletion and human toxicity, the environmental impact for the different scenarios studied is more or less of the same magnitude.

It is also noticeable when analyzing the data in Table 4 that the air based cooling technique is the most favorable one from an environmental aspect when compared to the other herein analyzed passive cooling techniques. The most influential environmental impact for all the cooling techniques is acidification as it can be seen in Fig. 29.

Finally, according to the provided general environmental analysis of the passive cooling techniques it was found that air based cooling technique has the lowest overall general impact to the environment (which is also additional advantage beside the LCOE aspect when compared to other analyzed passive cooling techniques). The PCM based cooling technique includes, compared to the reference PV system, the PCM material (paraffin based) and aluminum. This fact contributes without doubt, to more environmental impacts, global warming, acidification, eutrophication, photochemical ozone, in the construction phase of the LCA system studied (more materials, more complicated construction and installation phase consequently more environmental impacts to the construction phase).

The air based passive cooling technique also includes aluminum (less in quantity compared to PCM based cooling technique) and epoxy additives. This system has slightly more environmental impacts compared to PV reference system, due to the aluminum and epoxy glue, and less environmental impact compared to PCM based cooling technique (the air based passive cooling technique includes less quantity of aluminum compared to PCM based cooling technique).

Table 5
Environmental impact of different cooling techniques.

Environmental impact		PV-referent	PCM	Air based	Liquid based
Global Warming	tn CO_2 - eq	31.36518	53.74518	33.41563305	38.54518
Acidification	kg SO_2 - eq	123.9876	253.8876	135.3818944	141.0876
Eutrophication	kg PO4 - eq	18.88458	25.43958	19.4821003	20.48458
Ozone layer depletion	kg CFC - eq	0.00865458	0.010319	0.008795877	0.008671
Abiotic depletion	kg Sb - eq	23.9382	24.09029	23.96032181	23.9407
Human toxicity	tn Db - eq	17.5956	20.4326	18.10403706	17.9796
Photochemical ozone depletion	kg C_2H_4 - eq	6.77226	19.00226	7.443863553	10.62226

Finally, the liquid based passive cooling techniques includes in comparison to the other cooling techniques studied extra material for the basin construction. In our case the material used is polypropylene. Nevertheless, the environmental impact of the liquid based passive cooling technique is less significant in comparison to PCM based cooling technique.

6. Conclusions

A comprehensive review and analysis of passive cooling techniques for photovoltaic panels was carried out. It was found that the majority of the conducted research studies were focused on PCM based PV cooling, where most used Si-poly or Si-mono PV technologies and were examined in general. Further, it was also found that there is a general lack of field based studies as numerical and laboratory based studies are mostly represented. After PCM based passive cooling techniques, liquid based cooling techniques are the most considered ones, and are usually related to the liquid immersion method (water, nanofluids, etc.). Finally, air and radiative based passive cooling techniques are less investigated in current research studies. Furthermore, there is a lack of data in existing conducted research studies related to the analysis of general economic aspects for the proposed cooling techniques. The economic viability of the proposed cooling options is crucial for their potential market application, hence there is a significant gap related to that. The same issue was also noticed for LCA based analysis, i.e. the evaluation of environmental impacts for the considered cooling techniques, although one study was found that dealt with the LCA aspect of the specific case. Thus, it can be concluded that additional research efforts should be focused on in order to investigate both the economic and environmental aspects of the proposed passive cooling techniques for PVs.

Regarding the general performance analysis for the reviewed passive cooling techniques, it was found that efficiency increases can range just slightly above 1.0% to maximal 22% (peak value) for the liquid based immersion passive cooling method. However, it is certain that the value average related to the increased PV panel performance is significantly lower. Namely, for PCM based cooling, it is reasonable to expect an average increase in PV panel efficiency from 2% to 6%, from 10% to 15% for liquid based cooling, from 3% to 5% for air based cooling and between 0.5% to about 1.0% on average for radiative cooling. PCM based cooling techniques are more or less similar when compared to air based ones (with respect to the average increase in efficiency), however, PCM based cooling techniques are more expensive and complex to fabricate. Additional issues are related to the efficient thermal control (management) of PV/PCM configurations, cost issues and finally weight issues. Thus, the passive air based cooling technique emerges as the more favorable solution to adopt. The liquid based passive cooling option of PVs ensures the highest average increase in performance, however, it is not easy to implement in practical terms. There are issues related to the overall weight of the solution. Radiative base cooling techniques are less investigated and have less obtained efficiency increases, however, are still interesting for further research activity when compared to other passive cooling options.

A major contribution to the herein conducted research, besides the detailed review and evaluation of passive cooling techniques, is related to the general economic analysis of the considered passive cooling options. A 30 kW PV system case study was considered which operates in typical Mediterranean climate conditions, with available realistic data related to the investment, maintenance and finally annually gained electricity yield. Based on the provided LCOE analysis, it was found that depending from the considered passive cooling option, the LCOE ranges from 0.095 €/kW h to 0.185 €/kW h while for a non-cooled PV system, the LCOE is about 0.102 €/kW h. The LCOE aspect showed that the air based passive cooling technique for PVs is currently the most viable one, with an LCOE that ranges from 0.093 €/kW h to 0.095 €/kW h. Other considered passive techniques such as PCM and water either have a high LCOE or have technical difficulties for their application. The feasibility of radiative based cooling techniques was not analyzed due to rather modest efficiency increases. However, the PCM passive based cooling technique could be a realistic option, provided that a significant price drop of PCM materials was to occur or if novel PCM materials were to be developed. This can certainly be the case in the near future as part of a more advanced design of PV/PCM systems, which will ensure additional increases in efficiency, compared to existing PV/PCM passive cooling concepts. It is also important to emphasize that the conclusions drawn are based on, and apply to, PV systems for single and small scale applications. One can reasonably expect differences, when studying systems on a bigger scale, but that will be part of another research.

Regarding the environmental aspect, it was found that the most unfavorable herein analyzed cooling technique is currently the one with PCM based cooling. However, the air based cooling technique turned out to be the least harmful to the environment. In all the cases, the acidification impact to the environment was the most emphasized and was highest in magnitude.

Finally, based on the herein conducted analysis, the future development of passive based cooling techniques for PVs should focus on a smart integrated design of hybrid cooling techniques, where the integration of air and radiative based cooling techniques, ones with Al-fins mounted on the backside surface of PV panels, will be considered from the beginning. The point of a hybrid approach regarding more efficient passive cooling techniques is to emphasize the main individual advantage of each single passive cooling technique. It is also important to stress that there is still a demand for primary research data related to PV degradation, long term benefits, the impact on operation and maintenance and the longevity of the passive cooling systems in general. It is eventually long term performance that will determine the true LCOE of the PV systems and their place in the energy market as a result.

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